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NATIONAL SPACE ACADEMY

وكالة الإمارات للفضاء
UAE SPACE AGENCY

Introduction to Machine Learning

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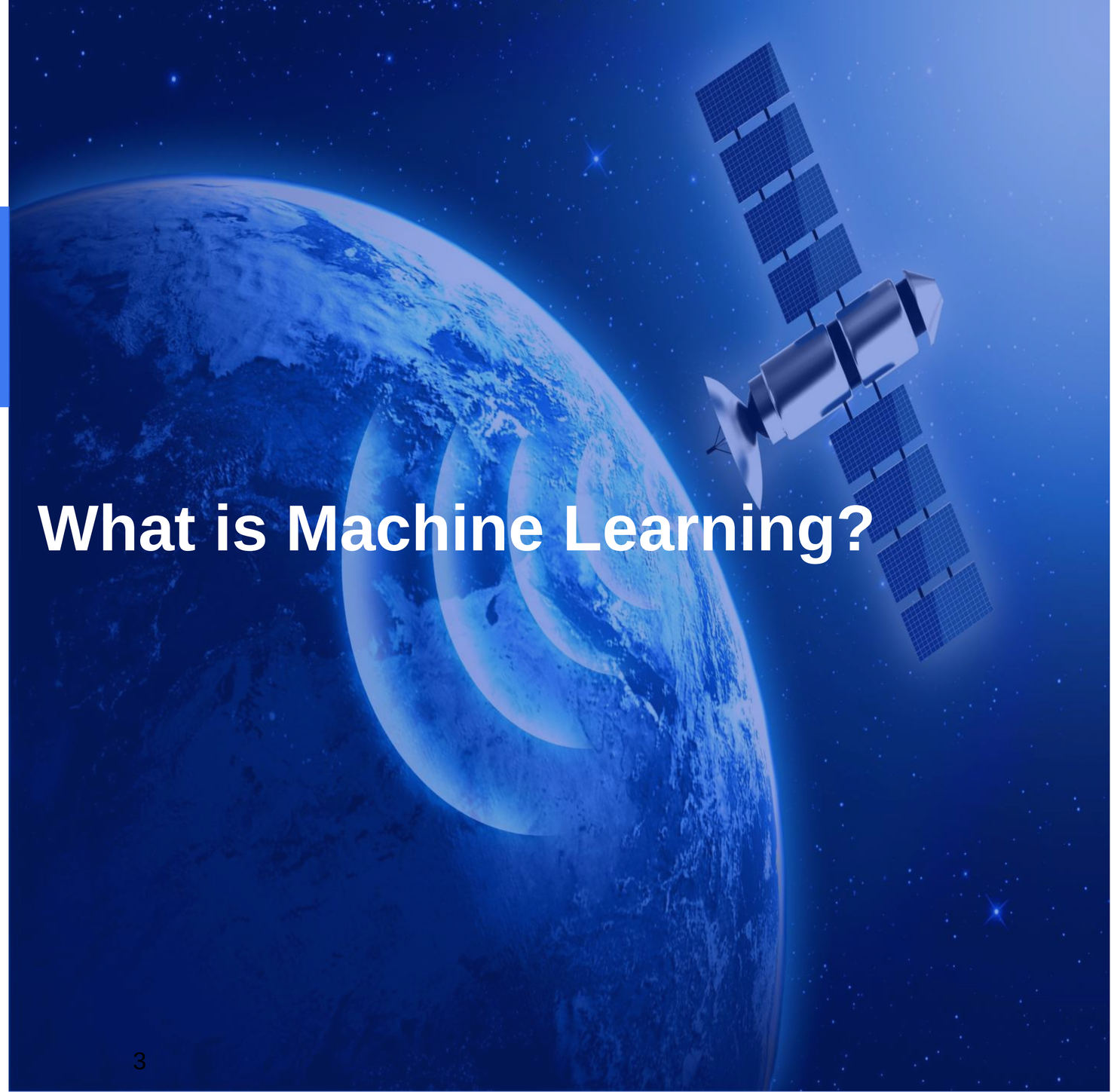
1. What is Machine Learning (ML)?

2. Machine Learning Fundamentals

3. Overview of Popular ML Models

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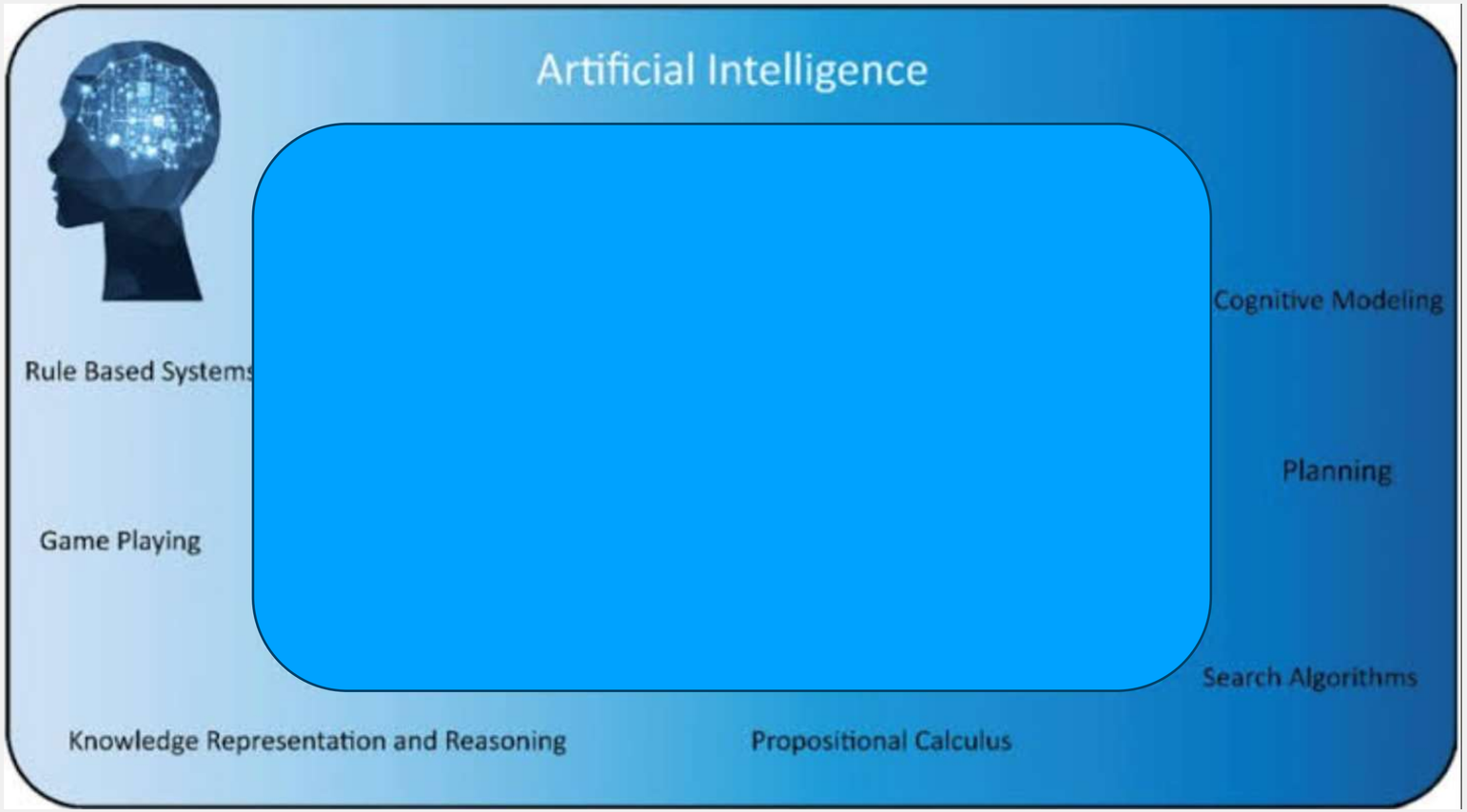
What is Machine Learning?



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1. What is Machine Learning

Artificial Intelligence (AI) is a broad field that refers to any technique that enables computers to mimic human intelligence — such as learning, problem-solving, or decision-making.

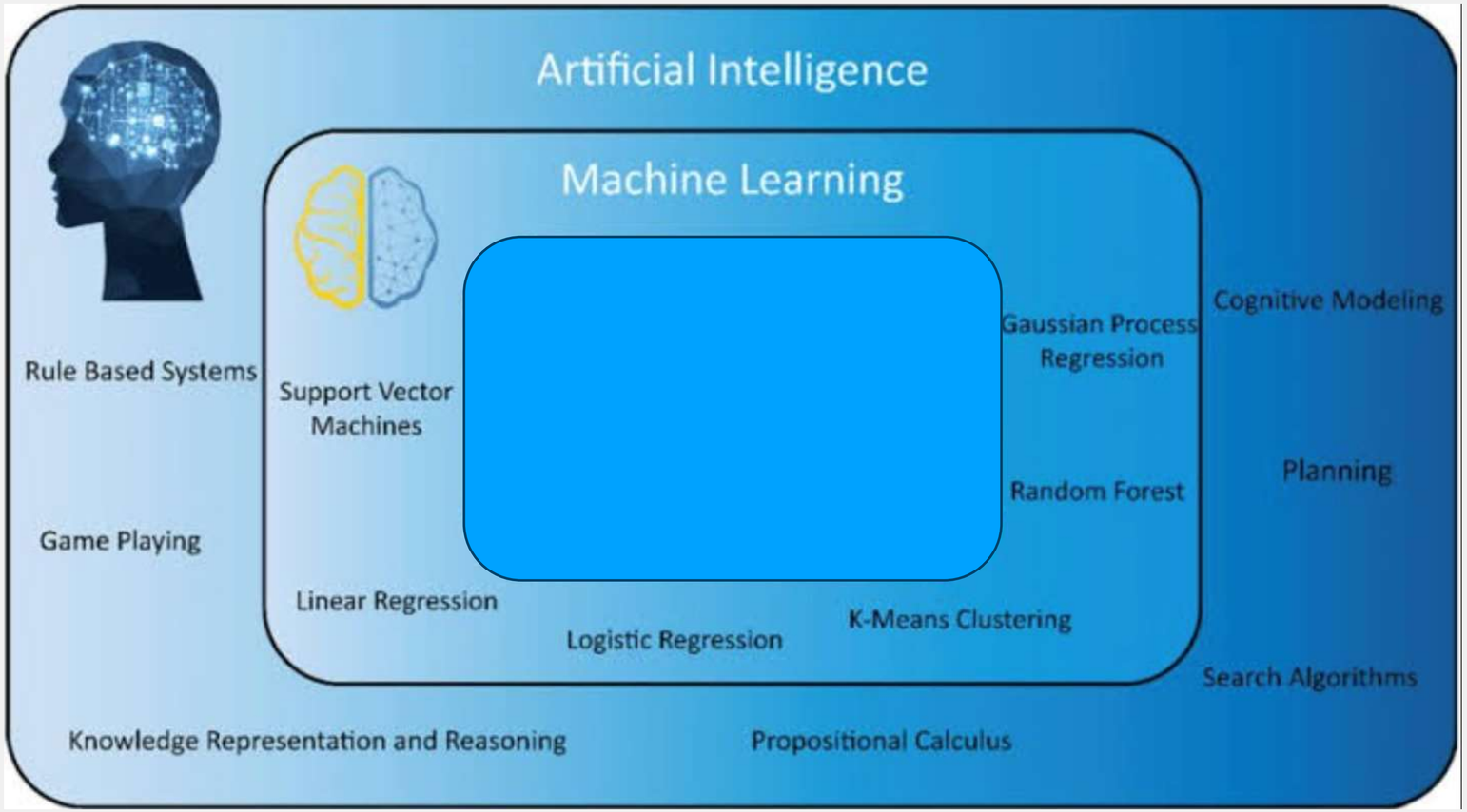


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- Machine Learning improves on traditional programming by allowing computers to **learn from data** and make decisions.
- It still requires **manual selection of features** to guide the learning process.

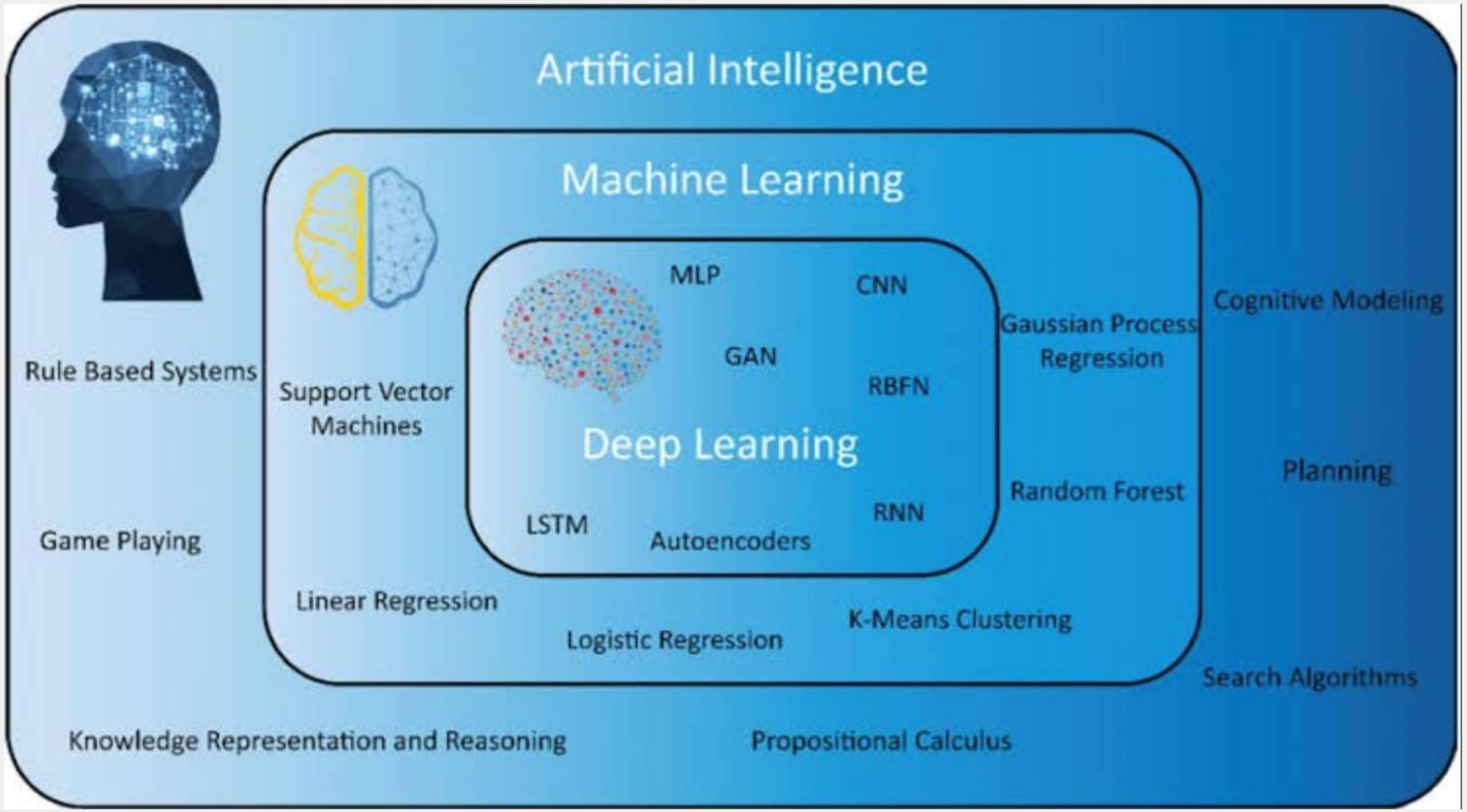


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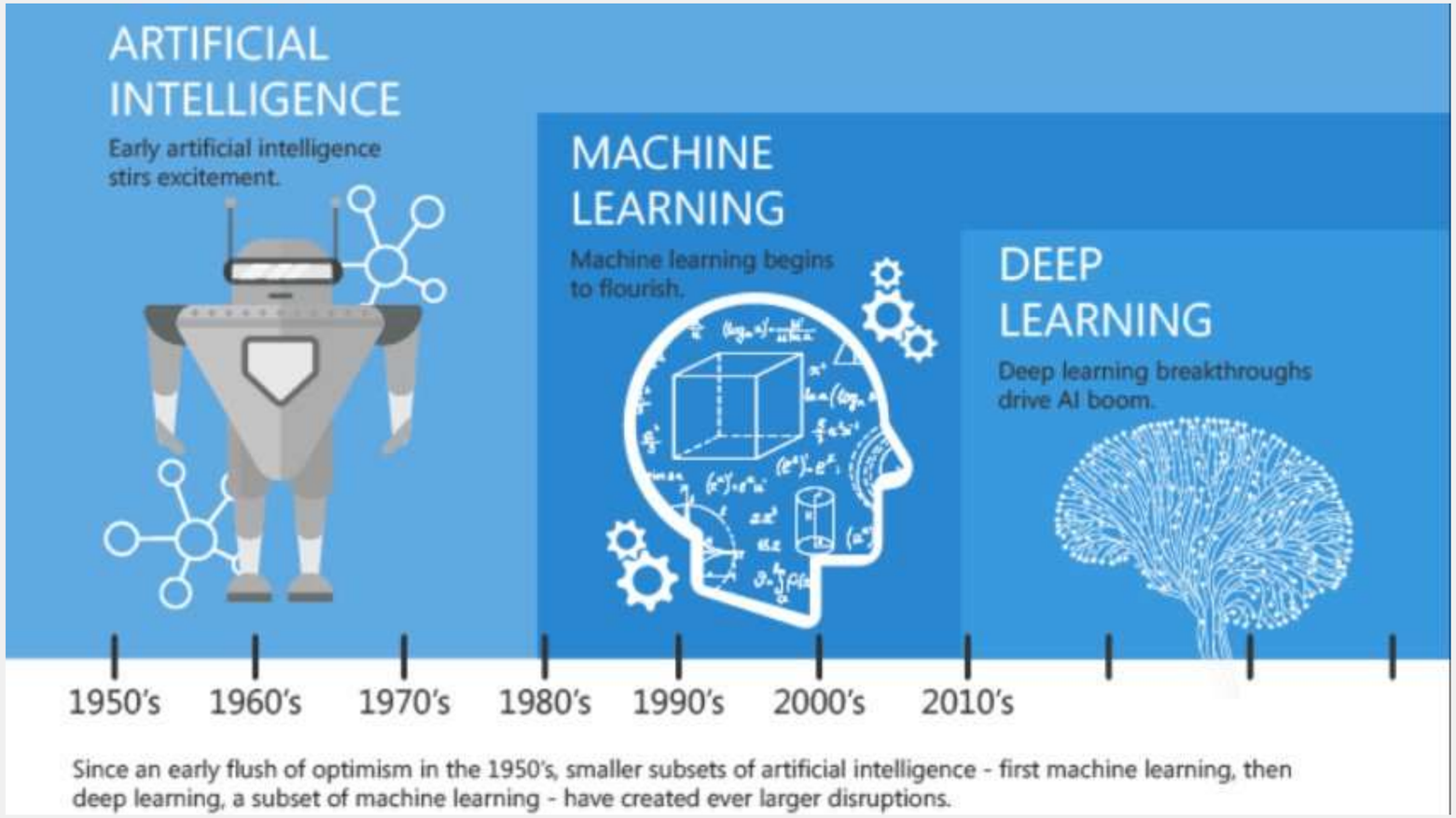
- Machine Learning improves on traditional programming by allowing computers to **learn from data** and make decisions.
- It still requires **manual selection of features** to guide the learning process.
- Deep Learning goes further by **automatically learning features** from raw data.
- It uses **neural networks** to detect complex patterns and make decisions.



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Timeline

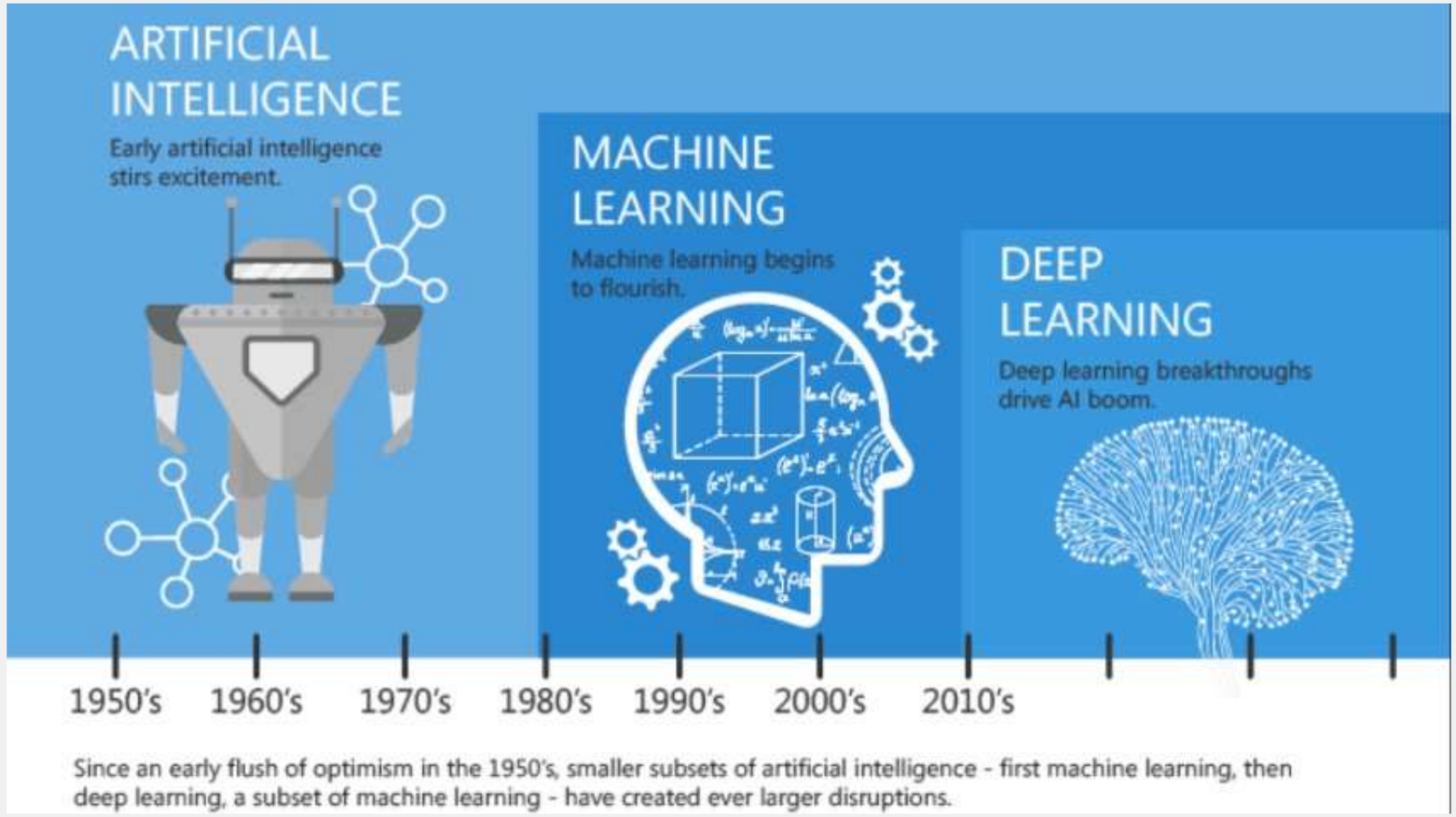
AI evolved through key milestones — rule-based systems in the early days, followed by Machine Learning as more data became available, and finally Deep Learning, which gained traction thanks to large datasets and powerful computing (like GPUs).

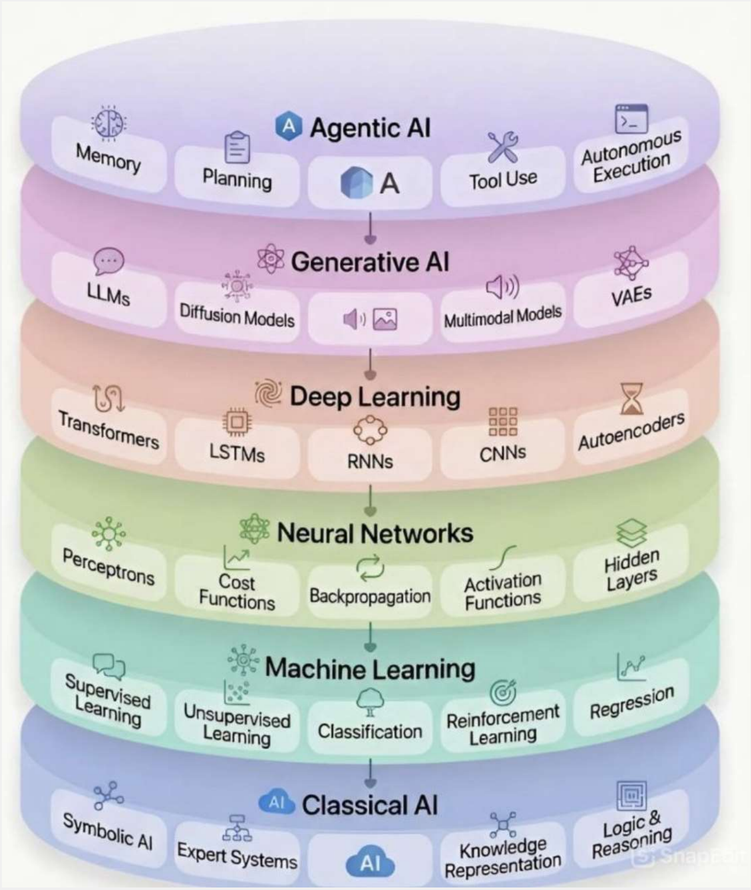


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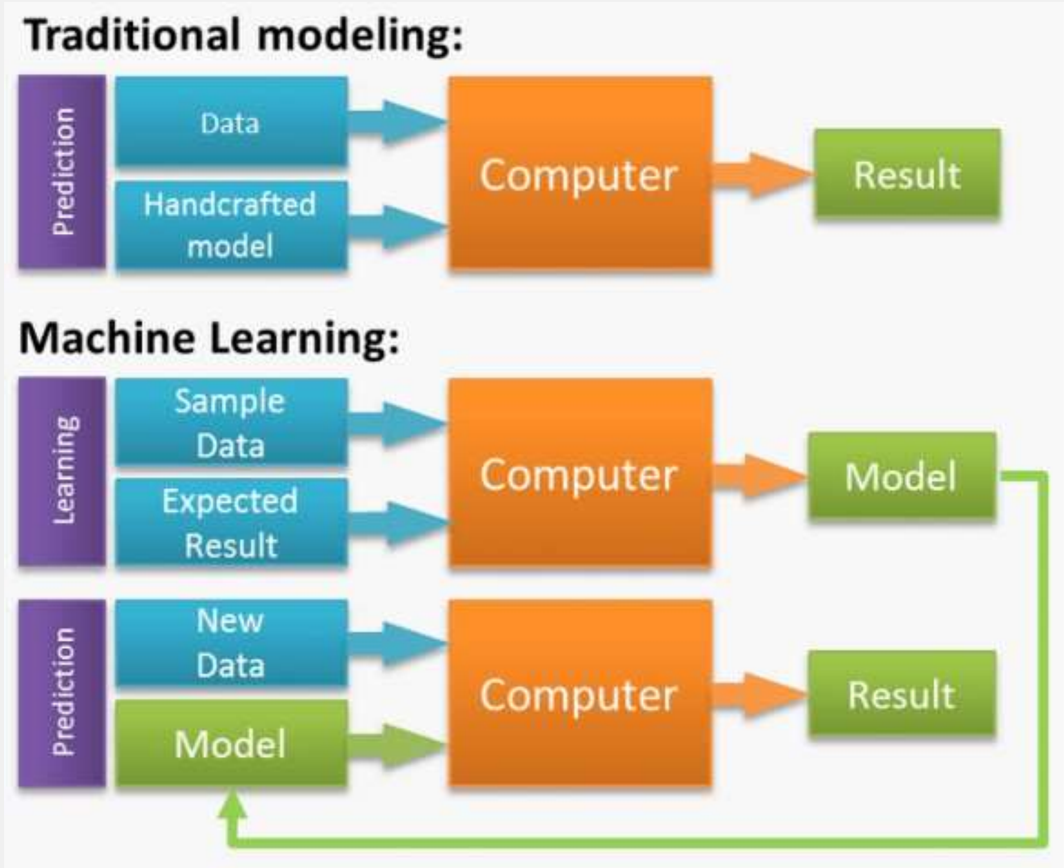
Classic Models vs Machine Learning

Traditional modelling

- The computer must be told exactly what to do
- The solution logic is explicitly programmed
- Struggles to handle new or incomplete data

Machine Learning/Deep Learning

- No need to manually define the solution
- The model learns to map inputs to outputs through training
- Can generalize to new situations and handle missing data



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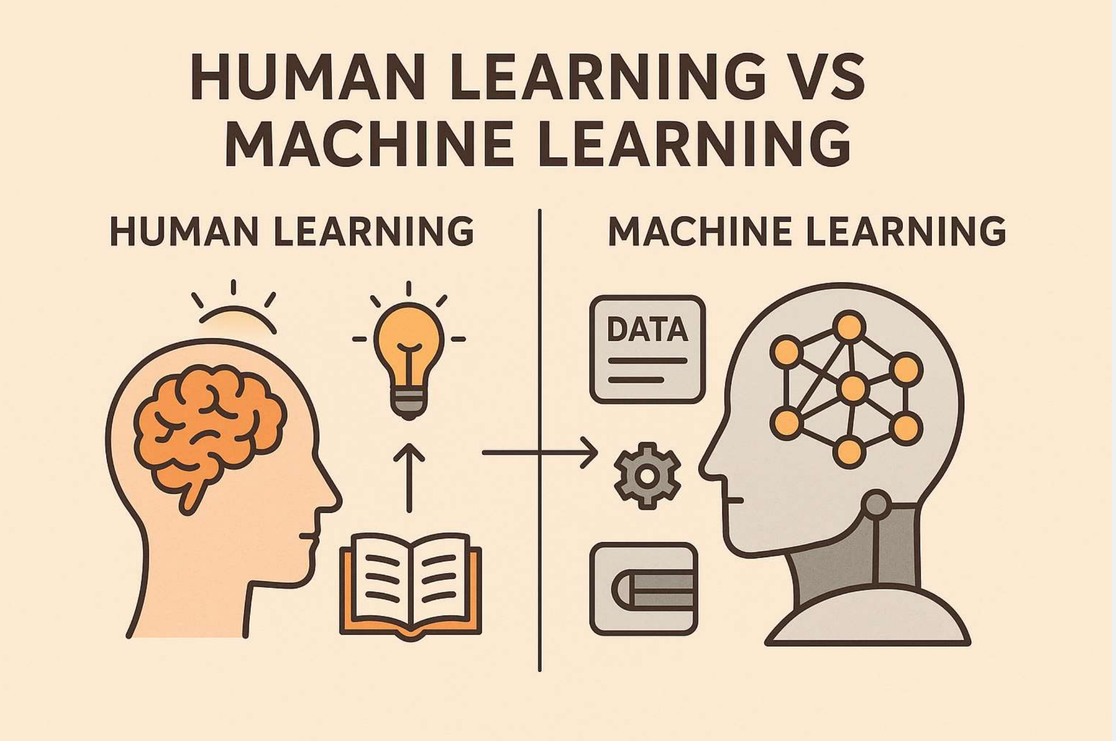
Human Learning vs Machine Learning

Human Learning

- We learn by **observing examples**, making mistakes, and improving over time
- Teachers provide **feedback** and we adjust our understanding
- We can **generalize** from past experiences to new situations

Machine Learning

- Machines learn from **large amounts of data** rather than explicit instructions
- A model is trained by comparing predictions to known outcomes and **adjusting itself**
- With enough data, it can generalize to **make predictions on new inputs**



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Human Vision vs Computer Vision

Human Vision

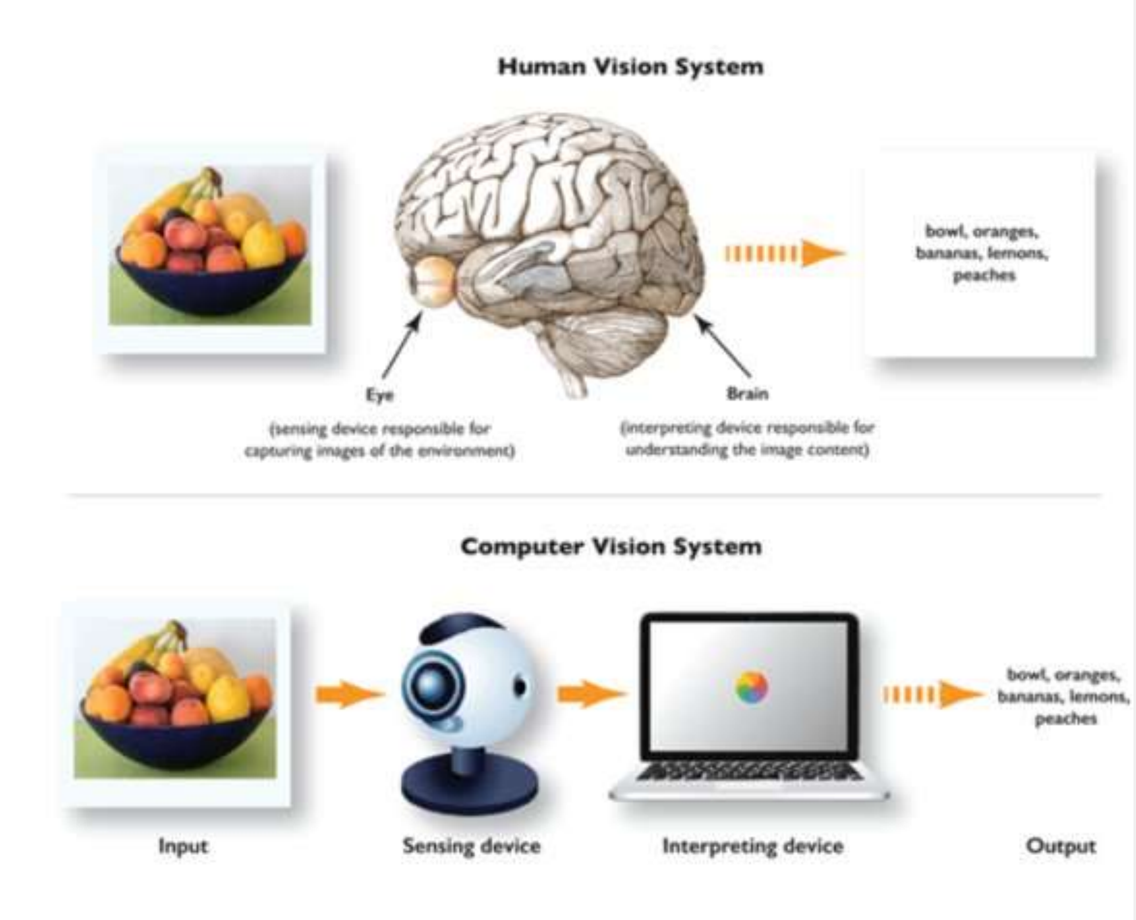
- The brain interprets visual information naturally, based on experience and context
- We recognize objects, patterns, and meanings intuitively

Computer Vision

Mimics how humans perceive the world, but through data and algorithms.

Main components:

- **Input:** Images or videos used for training
- **Processing:** Algorithms extract patterns and learn from the data
- **Output:** Predictions or classifications for new, unseen images



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Machine Learning in Everyday Life

- 🎬 **Streaming platforms** (YouTube, Netflix): Recommend content based on your viewing habits
- 📱 **Smartphones**: Unlock with face recognition or suggest auto-replies
- 🗣️ **Voice Assistants** (Siri, Alexa): Understand and respond to your voice
- 📧 **Email services**: Detect spam and filter messages
- 🌍 **Translation apps**: Convert text and speech between languages instantly
- 🤖 **ChatGPT and AI assistants**: Answer questions, assist with writing, and automate tasks using natural language understanding

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Machine Learning in Earth Observation

- **Land use detection**: Monitor urban growth, agriculture, and desert expansion
- **Vegetation analysis**: Track plant health and detect drought stress
- **Cloud detection**: Automatically identify and filter cloudy areas in satellite images
- **Environmental monitoring**: Detect oil spills or track marine pollution
- **Disaster response**: Assess damage after sandstorms or flooding using satellite data
- **Air quality monitoring**: Analyze satellite data to track pollution levels and support public health initiatives



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Machine Learning Fundamentals



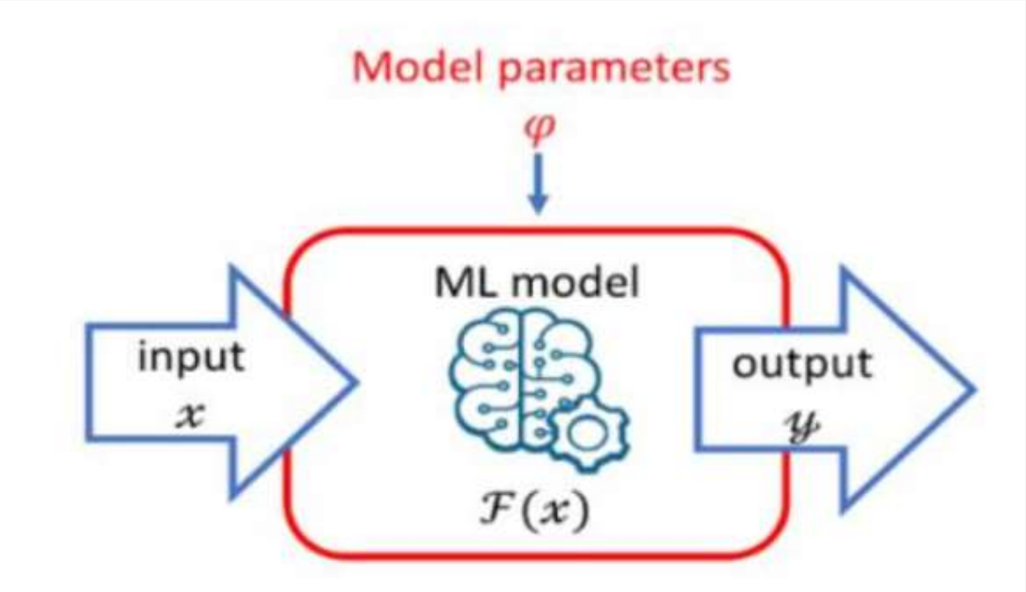
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What is a Machine Learning Model?

A machine learning model is a system that **learns to map inputs (X) to outputs (Y)** by finding patterns in data.

- You give it **examples**: input data (X) with the correct answers (Y)
- The model learns the relationship between them during **training**
- Once trained, it can **predict Y** for new, unseen X

Instead of programming the rules manually, we let the model **learn from data**.



ML model: $\mathcal{F}_{parameters}(x; weights)$

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Types of Machine Learning

Supervised Learning
Learns from labeled data to predict outcomes or classify new data.
X ➔ model(X) ➔ Y

Unsupervised Learning
Finds patterns or groups in unlabeled data without predefined answers.
X ➔ model(X) ➔ patterns in X

Reinforcement Learning
Learns by trial and error, receiving feedback through rewards or penalties.
X (state) ⇒ Agent/Model ⇒ Y (action)

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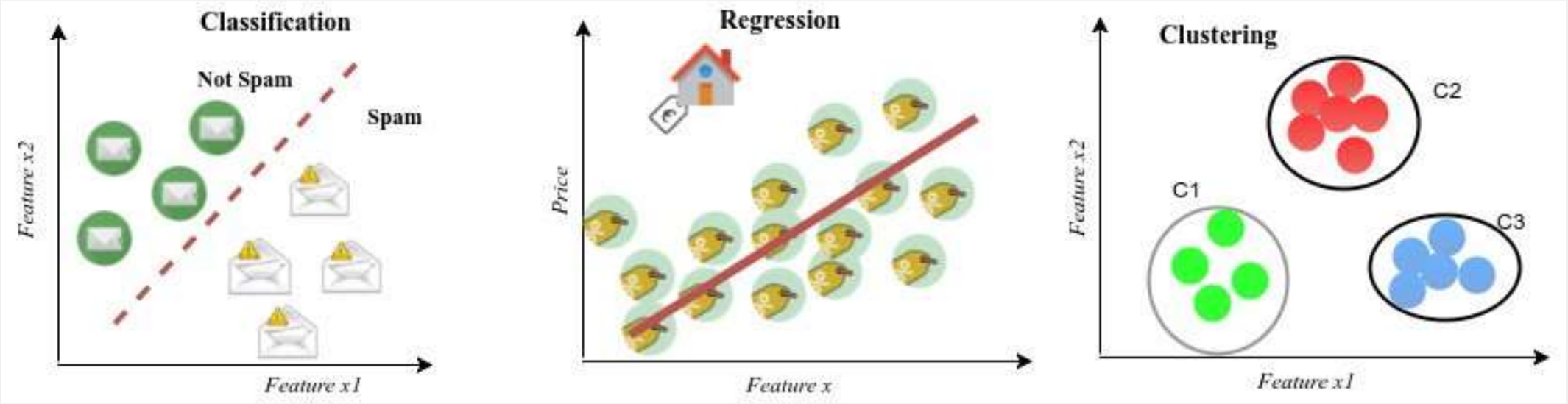
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Classification: Predicts discrete categories or classes.

Regression: Predicts continuous numerical values.

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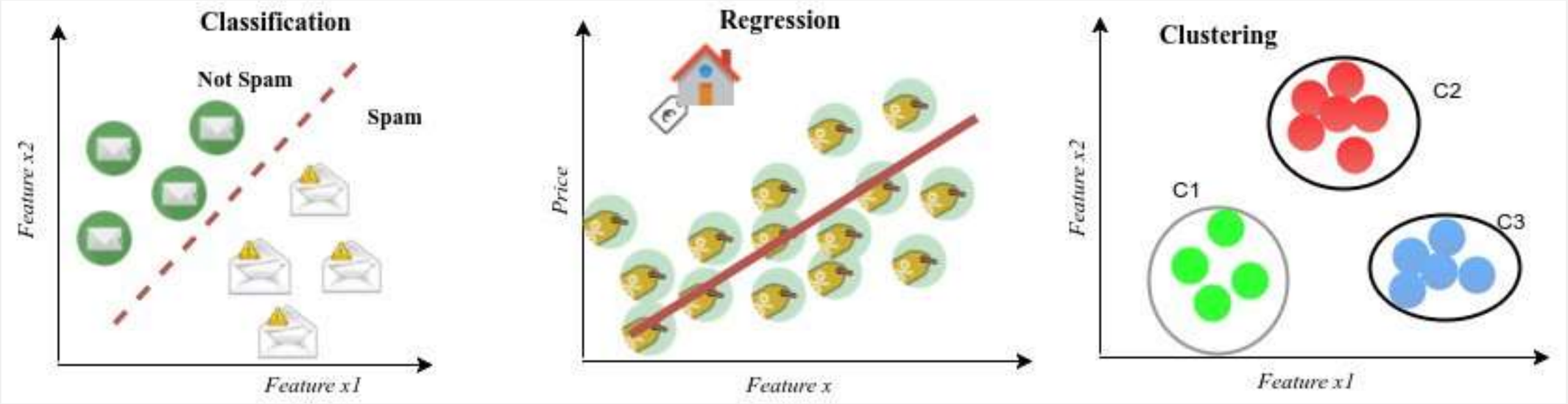
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Regression: Predicts continuous numerical values.

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Clustering: Groups similar data points together without predefined labels.



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The Machine Learning Pipeline

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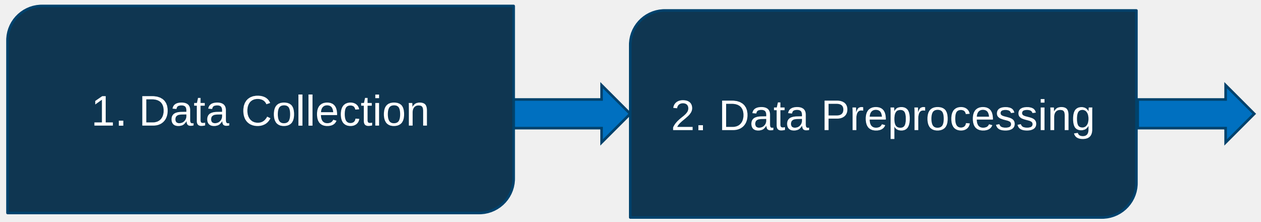
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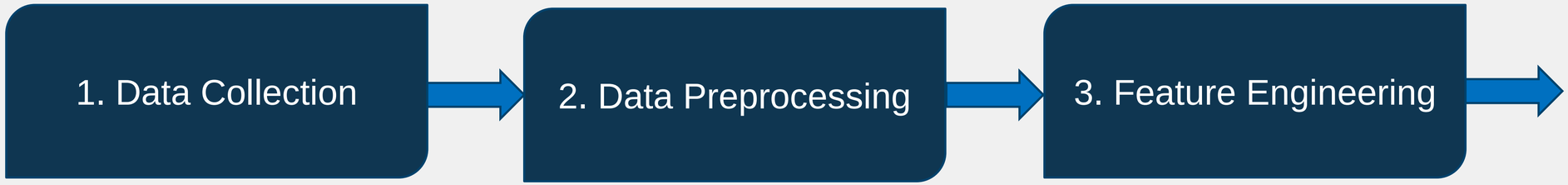


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2. Data Preprocessing: Clean, format and prepare data (handle missing value, normalize, encode categorical data, etc).

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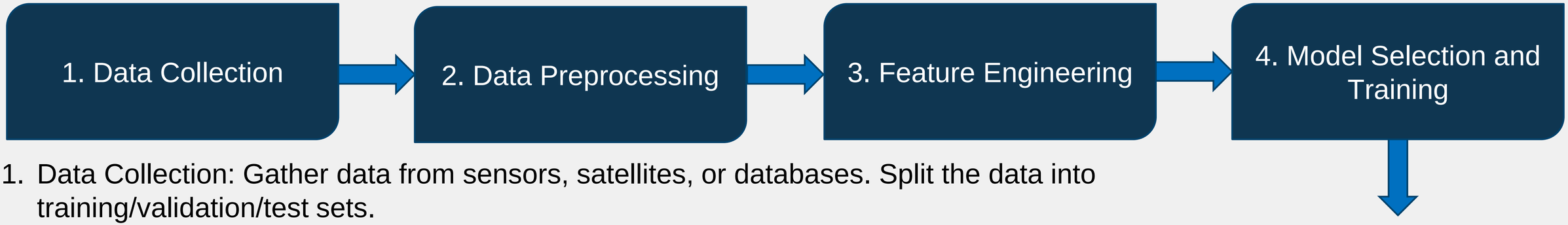
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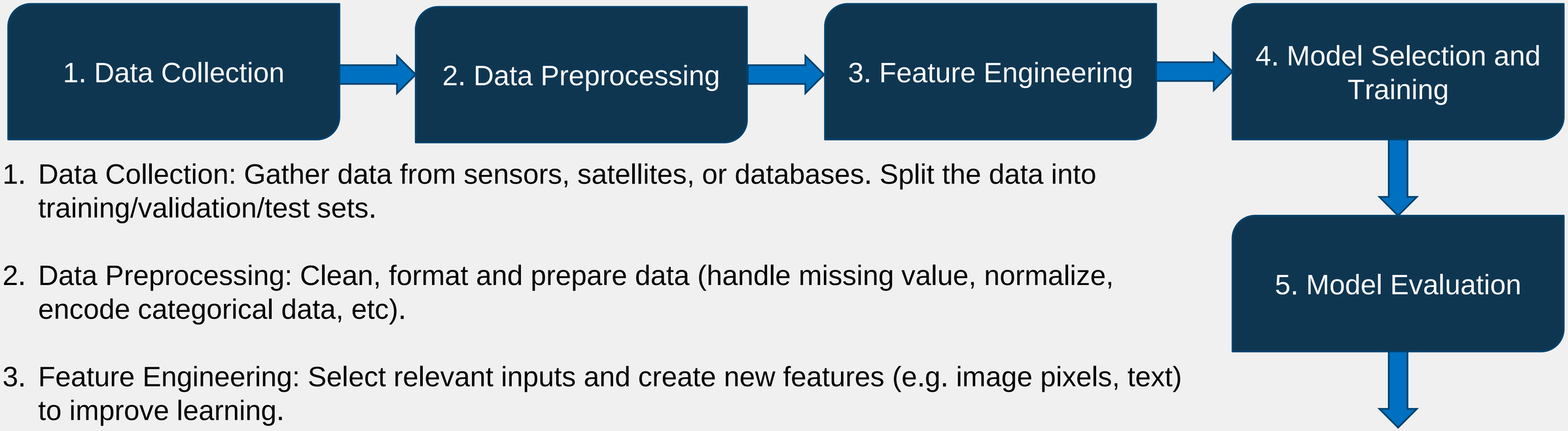
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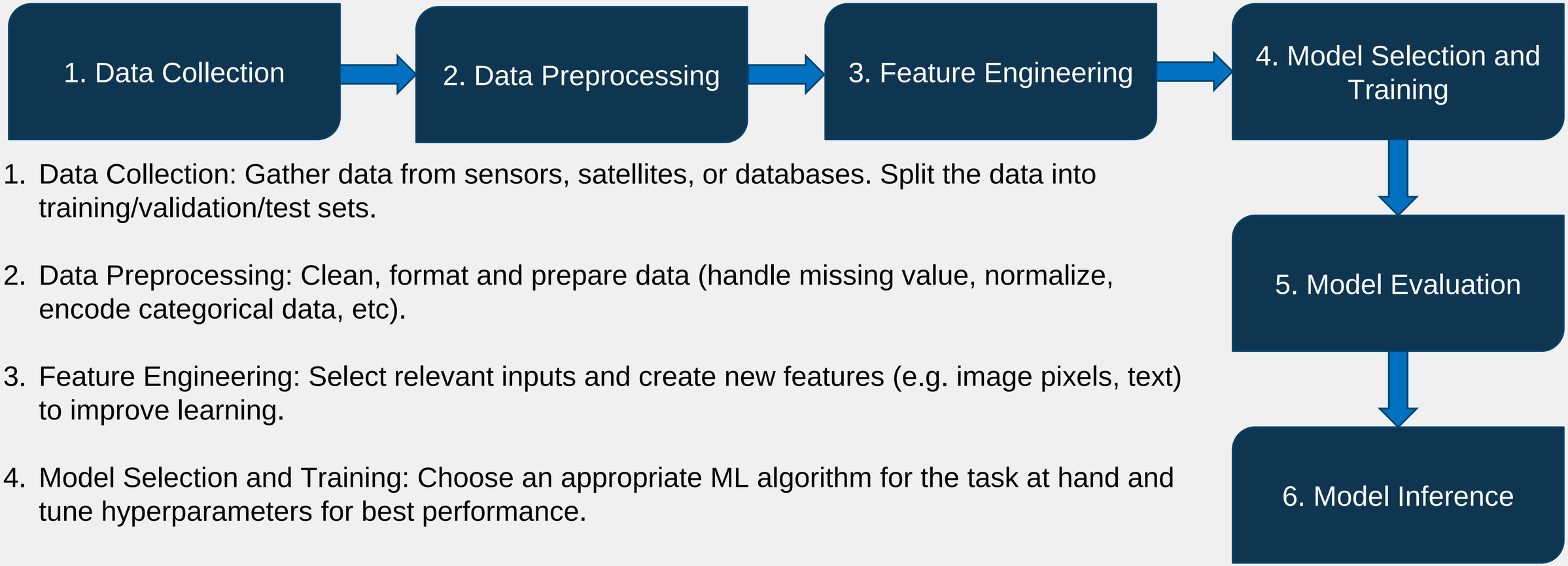
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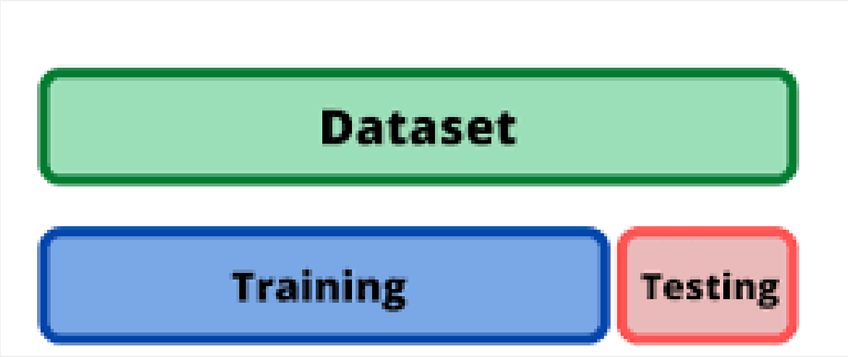
5. Model Evaluation: Assess the model's accuracy and generalization using test data.

6. Model Inference: Use the trained model to make prediction and gain insights from new data.

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Train-Validation-Test Split

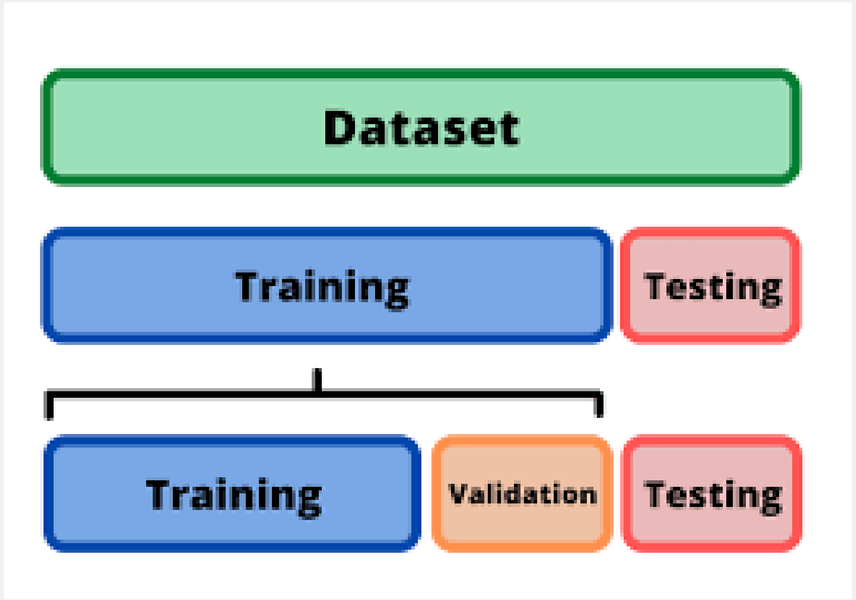
- **Train-test split** separates the data for training the model and evaluating it on unseen examples.
- **Training set** is used to fit the model; **test set** is for final performance check.



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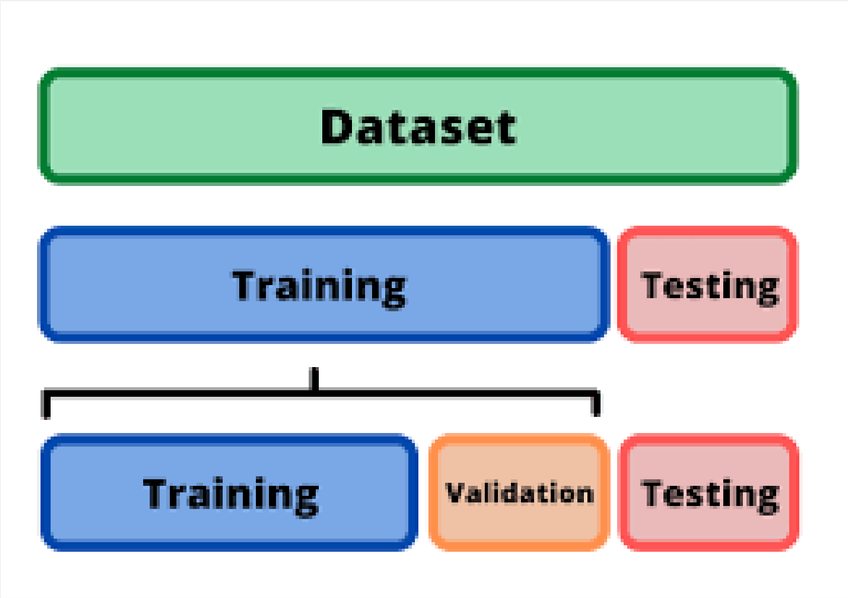


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Train-Validation-Test Split

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- **Validation set** (optional) helps tune the model (e.g., during hyperparameter optimization).
- Use shuffle=True to avoid order bias — ⚠ **Not recommended** for **time series** or **spatial data**, where order or location matters.
- For classification, use stratify=Y to keep the same class proportions in each split

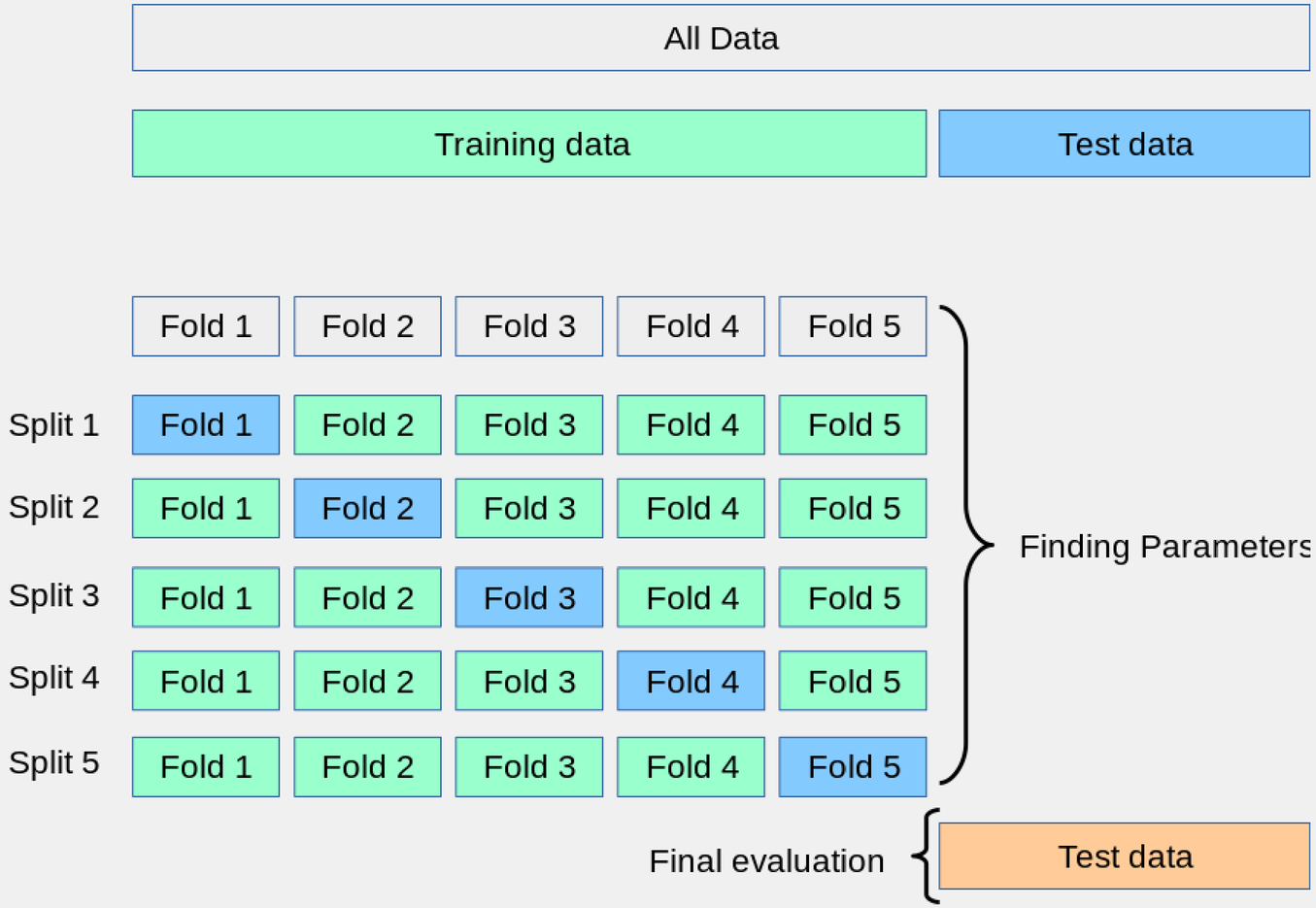
https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.train_test_split.html



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K-fold Cross-Validation

- **Cross-validation** helps evaluate a model's performance more reliably using only the training data.
- In **k-fold CV**, the training set is split into *k* parts — the model trains on *k−1* and validates on the remaining fold, repeated *k* times.
- This reduces the risk of biased evaluation due to a "lucky" or "unlucky" split.
- Final performance is averaged over the *k* folds.
- Common choice: **k = 5** or **10**.
- The **test set remains untouched** until final evaluation.



https://scikit-learn.org/stable/modules/cross_validation.html

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Simplest Regression Model: Linear Regression (single feature case)

◆ **What it does**

•Linear Regression models the relationship between input features **X** and a continuous output **y**.

•The model tries to find the **best-fitting line** (or hyperplane) through the data.

◆ **Prediction Formula**

$$y = w_1 x + b_1$$

•Where:

- x = input feature
- w_1 = slope
- b_1 = intercept
- y = predicted output or target

◆ **Training the Model**

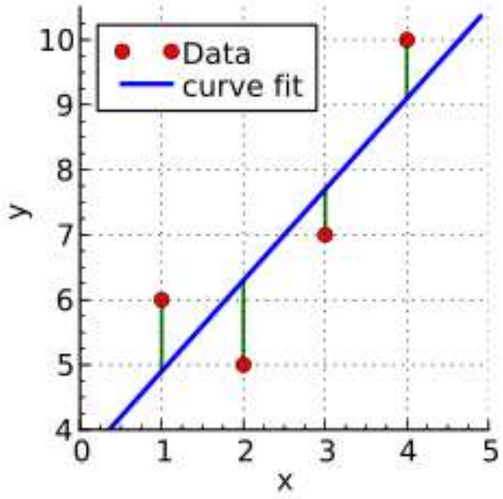
•The model learns **weights (w)** by minimizing the **Mean Squared Error (MSE)** between predictions and true values

•This is typically done using optimization methods like **Ordinary Least Squares (OLS)** or **gradient descent**.

◆ **Outputs of the Model**

•Once trained, the model provides:

- Predictions $\hat{y}$ for new data.
- Performance metrics like **(R)MSE**, **MAE**, **R² score**, to evaluate accuracy.



$$MAE = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}|$$

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2}$$

$$R^2 = 1 - \frac{\sum(y_i - \hat{y})^2}{\sum(y_i - \bar{y})^2}$$

Where,

$\hat{y}$ – *predicted value of* y

$\bar{y}$ – *mean value of* y

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Simplest Regression Model: Linear Regression (multiple features)

◆ **What it does**

•Linear Regression models the relationship between input features **X** and a continuous output **y**.

•The model tries to find the **best-fitting line** (or hyperplane) through the data.

◆ **Prediction Formula**

$$y = w_1 x_1 + b_1 + w_2 x_2 + b_2 + w_3 x_3 + b_3 + \ ...$$

•Where:

- x_i = input features
- w_i = **weights**
- b_i = **biases**
- y = predicted output or target

◆ **Training the Model**

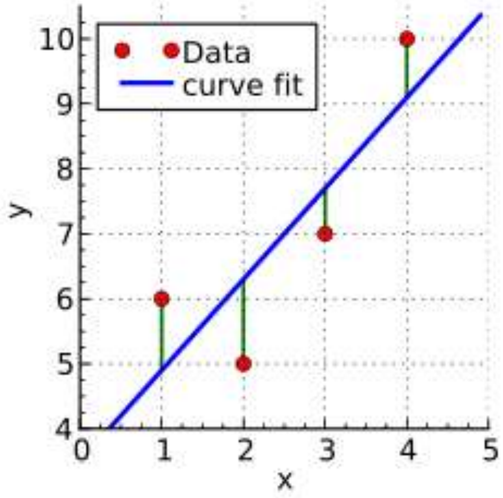
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Simplest Regression Model: Linear Regression

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•The model tries to find the **best-fitting line** (or hyperplane) through the data.

◆ **Prediction Formula**

$$y = WX + B$$

•Where:

- X = input features in vector format
- W = **weights** in vector format
- B = **biases** in vector format
- y = predicted output or target

◆ **Training the Model**

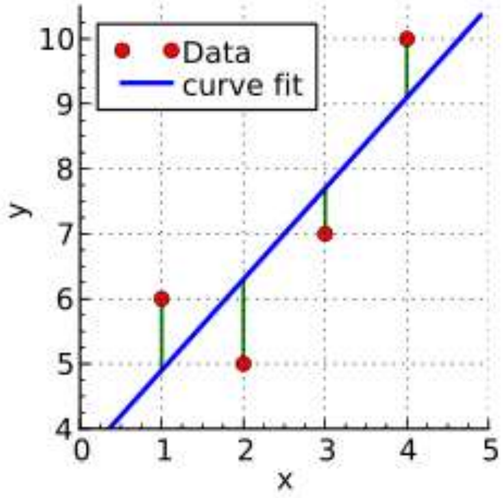
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$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2}$$

$$R^2 = 1 - \frac{\sum(y_i - \hat{y})^2}{\sum(y_i - \bar{y})^2}$$

Where,

$\hat{y}$ – *predicted value of* y

$\bar{y}$ – *mean value of* y

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Underfitting vs Overfitting

🔹 What is a Loss Function?

It measures how far off the model's predictions are from the actual values.
The goal is to **minimize this error** during training.
Example

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

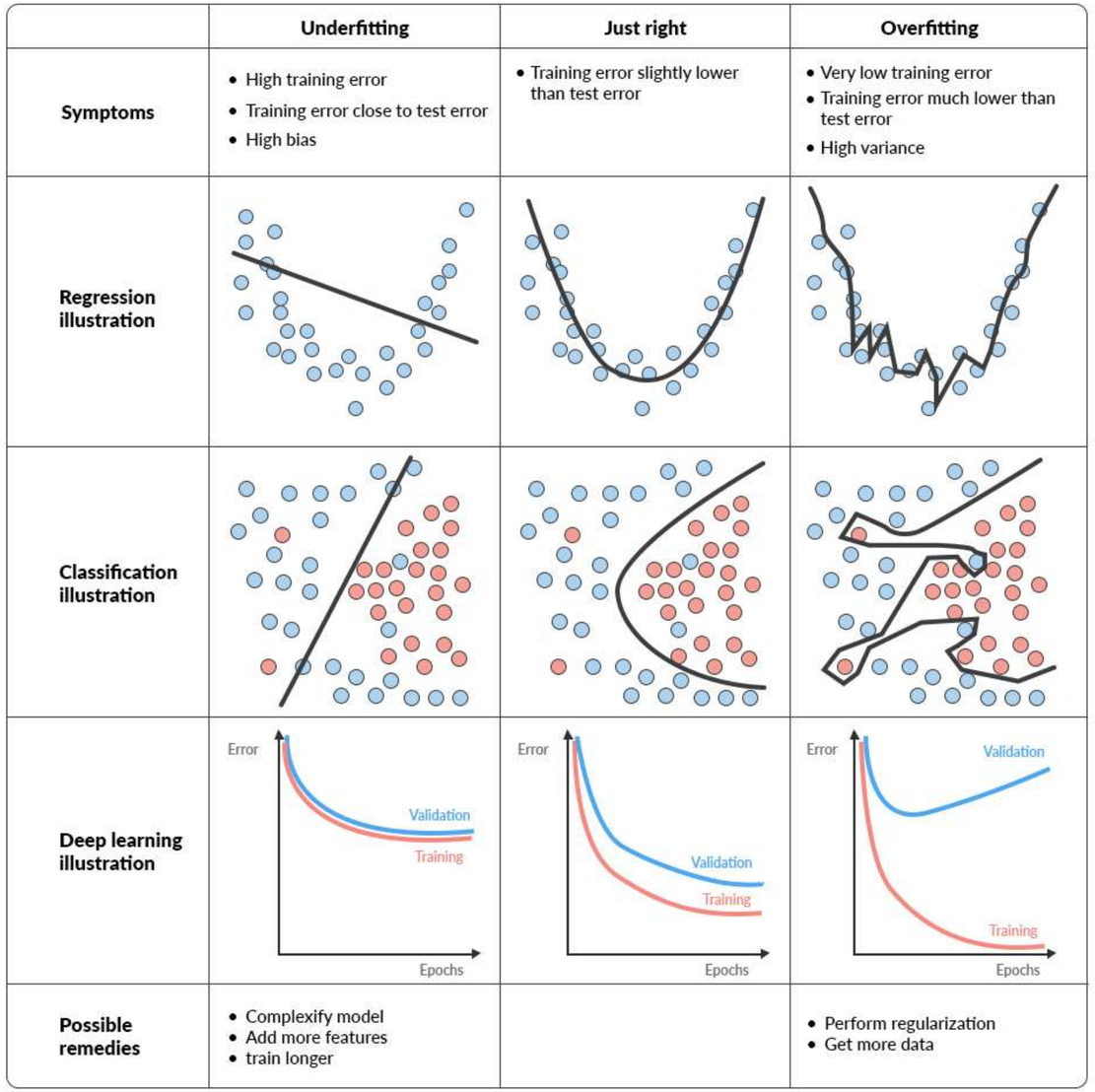
🔹 Fitting the Model

The training process searches for model weights that **minimize the loss** using techniques like **gradient descent**.

🔹 Underfitting vs Overfitting

•**Underfitting**: The model is too simple, fails to capture the data pattern.
•**Overfitting**: The model is too complex, memorizes training data but fails on new data.
✅ The sweet spot: **Good fit** with strong generalization to unseen examples.

Additional resources: https://uniathena.com/understanding-bias-variance-tradeoff-balance-model-performance



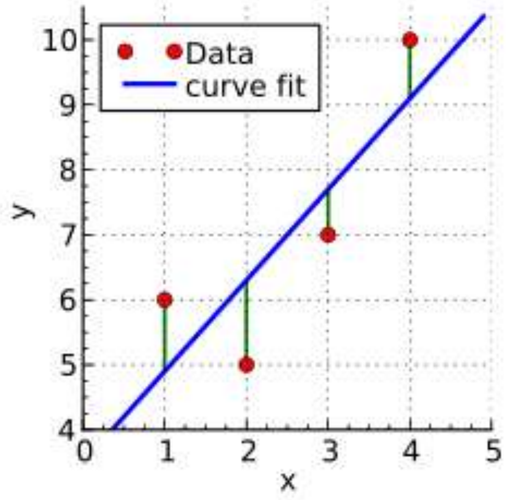
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Regression Metrics

Any regression model can be evaluated using the same metrics but with the predictions

$$\hat{y} = \mathcal{F}_{parameters}(X; W)$$

- **Mean Absolute Error (MAE)**: The average of the absolute errors between the predicted and actual values.
- **Mean Squared Error (MSE)**: The average of the squared errors between the predicted and actual values.
- **Root Mean Squared Error (RMSE)**: The square root of MSE.
- **R-squared (R^2)**: The proportion of the variance in the dependent variable that is predictable from the independent variables.



$$MAE = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}|$$

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2}$$

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Where,

$\hat{y}$ – *predicted value of y*

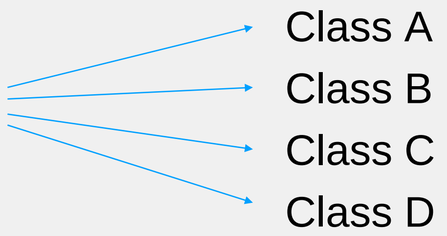
$\bar{y}$ – *mean value of y*

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Classification Metrics: Accuracy

- Classification model:

$$\hat{y} = \mathcal{F}_{parameters}(X; W) \Rightarrow \text{discrete}$$



- Accuracy metric:

$$Accuracy = \frac{Number\ of\ correct\ predictions}{Total\ number\ of\ predictions}$$

Limitations of Accuracy

Can be misleading when the target classes are imbalanced.

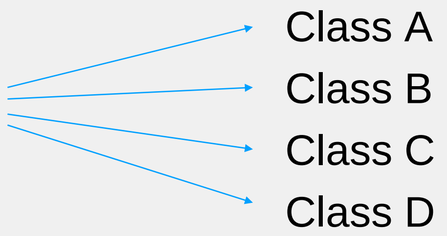
- Example: If 95% of the data is class A, a model always predicting A gets 95% accuracy — but it's not useful.

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Classification Metrics: Confusion Matrix

- Classification model:

$$\hat{y} = \mathcal{F}_{parameters}(X; W) \Rightarrow \text{discrete}$$



	Actual Class A	Actual Class B	Actual Class C	Actual Class D
Predicted Class A	102	3	0	4
Predicted Class B	3	23	2	1
Predicted Class C	0	5	10	4
Predicted Class D	2	0	0	2

- In multi-class settings, confusion matrices can become **hard to interpret**, especially with many classes — they show errors but not how **severe** or **important** each mistake is.

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Binary Classification Metrics: Accuracy

- Classification model:

$$\hat{y} = \mathcal{F}_{parameters}(X; W) \Rightarrow \text{discrete}$$

Class A ➔ 0

Class B ➔ 1

- Some models can only predict 0 or 1 natively, but some other models can predict a value between 0 and 1 that can be interpreted as the probability to be in class A or B.
- The threshold to go from the probability prediction to the predicted classes is set to 0.5 by default but can be adapted.
- Accuracy metric:

$$Accuracy = \frac{Number\ of\ correct\ predictions}{Total\ number\ of\ predictions}$$

Limitations of Accuracy

- Can be misleading when the target classes are imbalanced.
- Does not take into account the actual probability prediction but only the prediction after applying a threshold.

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Binary Classification Metrics: Confusion Matrix

- Classification model:

$$\hat{y} = \mathcal{F}_{parameters}(X; W) \Rightarrow \text{discrete}$$

Class A ➔ 0 ➔ Negative

Class B ➔ 1 ➔ Positive

	Actual Positive Class	Actual Negative Class
Predicted Positive Class	True Positives (TP)	False Positives (FP)
Predicted Negative Class	False Negatives (FN)	True Negatives (TN)

- Total P = TP + FN
- Total N = FP + TN

- TP (True positive) : Actual class is positive and the model predicted as positive
- FN (False negative) : Actual class is positive **but** the model predicted as negative
- TN (True negative) : Actual class is negative and the model predicted as negative
- FP (False positive) : Actual class is negative **but** the model predicted as positive

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Binary Classification Metrics: Confusion Matrix Derivatives

- Classification model:

$$\hat{y} = \mathcal{F}_{parameters}(X; W) \Rightarrow \text{discrete}$$

Class A ➔ 0 ➔ Negative

Class B ➔ 1 ➔ Positive

	Actual Positive Class	Actual Negative Class
Predicted Positive Class	True Positives (TP)	False Positives (FP)
Predicted Negative Class	False Negatives (FN)	True Negatives (TN)

Metric	Formula	Interpretation
Accuracy	$\frac{TP + TN}{TP + TN + FP + FN}$	Overall performance of model
Precision	$\frac{TP}{TP + FP}$	How accurate the positive predictions are
Recall Sensitivity	$\frac{TP}{TP + FN}$	Coverage of actual positive sample
Specificity	$\frac{TN}{TN + FP}$	Coverage of actual negative sample
F1 score	$\frac{2TP}{2TP + FP + FN}$	Hybrid metric useful for unbalanced classes

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Binary Classification Metrics: Precision, Recall and F1-score

	Actual Positive Class	Actual Negative Class
Predicted Positive Class	TP	FP
Predicted Negative Class	FN	TN

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall\ (TPR) = \frac{TP}{TP + FN}$$

$$False\ Positive\ Rate\ \ (TPR) = \frac{FP}{FP + TN}$$

$$F1\ score = \frac{2 \times precision \times recall}{precision + recall}$$

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Binary Classification Metrics: Example

	Actual Positive Class	Actual Negative Class
Predicted Positive Class	TP = 1	FP = 0
Predicted Negative Class	FN = 8	TN = 49991

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall\ (TPR) = \frac{TP}{TP + FN}$$

$$False\ Positive\ Rate\ \ (TPR) = \frac{FP}{FP + TN}$$

$$F1\ score = \frac{2 \times precision \times recall}{precision + recall}$$

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Binary Classification Metrics: Example

	Actual Positive Class	Actual Negative Class
Predicted Positive Class	TP = 1	FP = 0
Predicted Negative Class	FN = 8	TN = 49991

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} = 99.9\%$$

$$Precision = \frac{TP}{TP + FP} = 100\%$$

$$Recall\ (TPR) = \frac{TP}{TP + FN} = 11.11\%$$

$$False\ Positive\ Rate\ \ (TPR) = \frac{FP}{FP + TN} = 0\%$$

$$F1\ score = \frac{2 \times precision \times recall}{precision + recall} = 20\%$$

- Relevant for imbalanced classes
- Still defined for a single threshold
  ➔ same value independently of the actual probability predicted

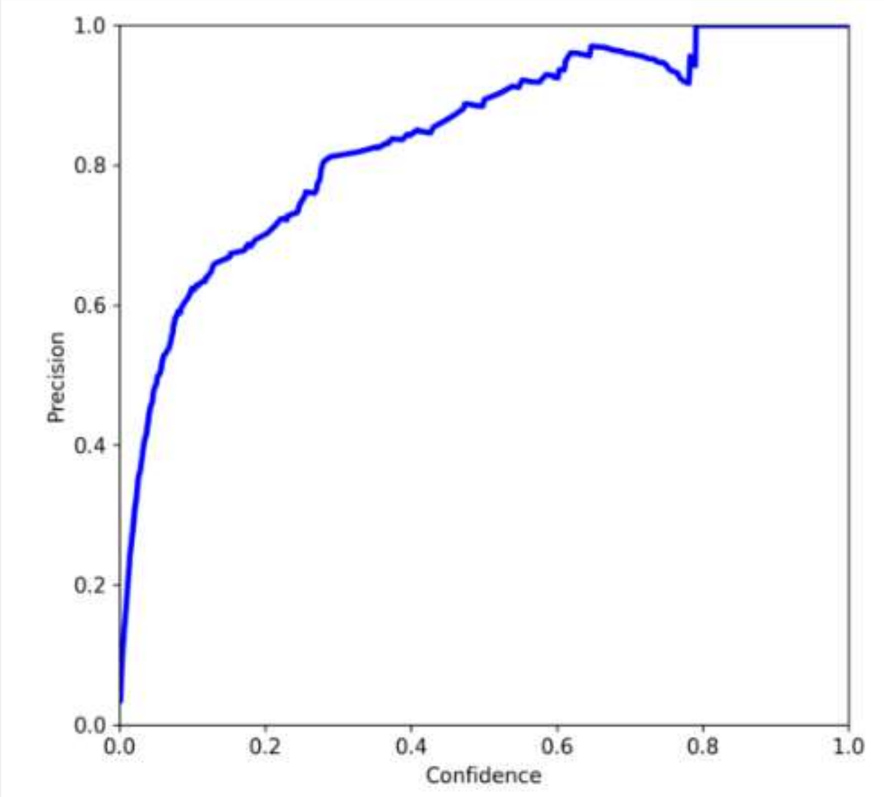
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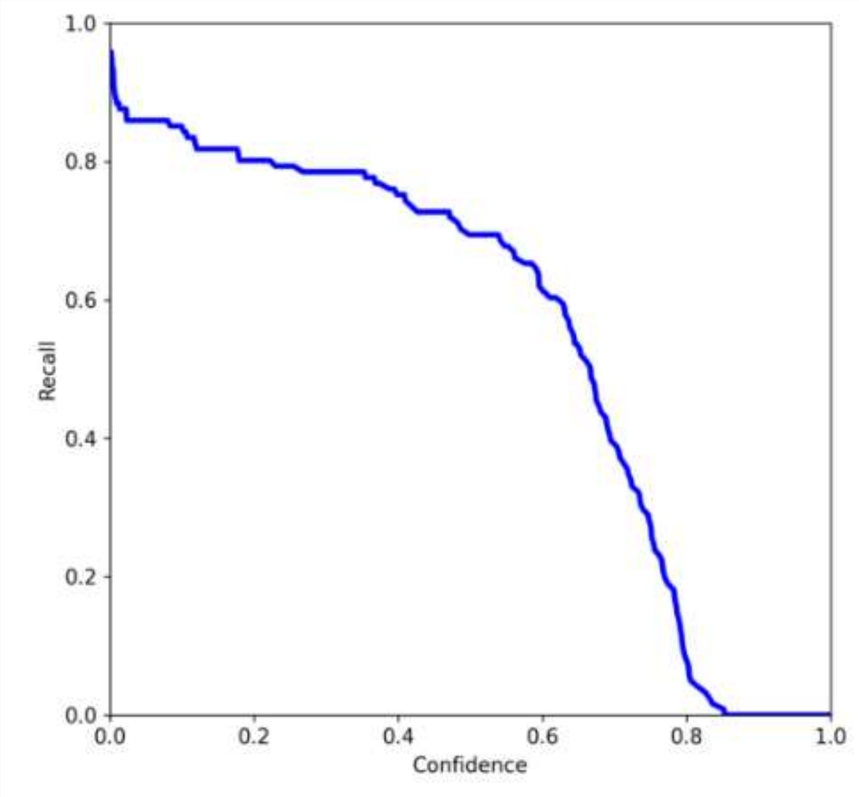
Binary Classification Metrics: Precision-Recall Curve, AUC and AP

$$Precision = \frac{TP}{TP + FP}$$

	Actual Positive Class	Actual Negative Class
Predicted Positive Class	TP	FP
Predicted Negative Class	FN	TN

$$Recall = \frac{TP}{TP + FN}$$





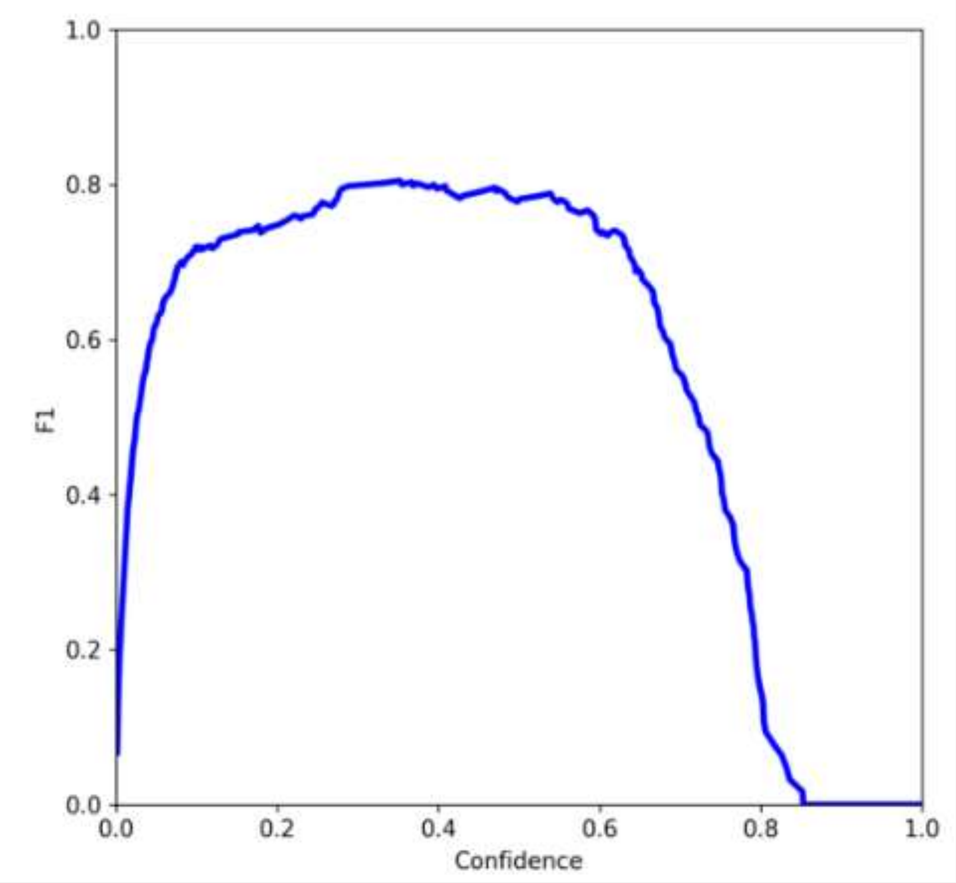
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Binary Classification Metrics: Precision-Recall Curve, AUC and AP

$$Precision = \frac{TP}{TP + FP}$$

	Actual Positive Class	Actual Negative Class
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$$Recall = \frac{TP}{TP + FN}$$



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Binary Classification Metrics: Precision-Recall Curve, AUC and AP

$$Precision = \frac{TP}{TP + FP}$$

	Actual Positive Class	Actual Negative Class
Predicted Positive Class	TP	FP
Predicted Negative Class	FN	TN

$$Recall = \frac{TP}{TP + FN}$$

◆ **Precision-Recall Curve**

•Shows the trade-off between **precision** and **recall** at different classification thresholds.

•Useful when classes are **imbalanced**, and **positive class** is more important.

◆ **Area Under Curve (AUC-PR)**

•The **area under the precision-recall curve**.

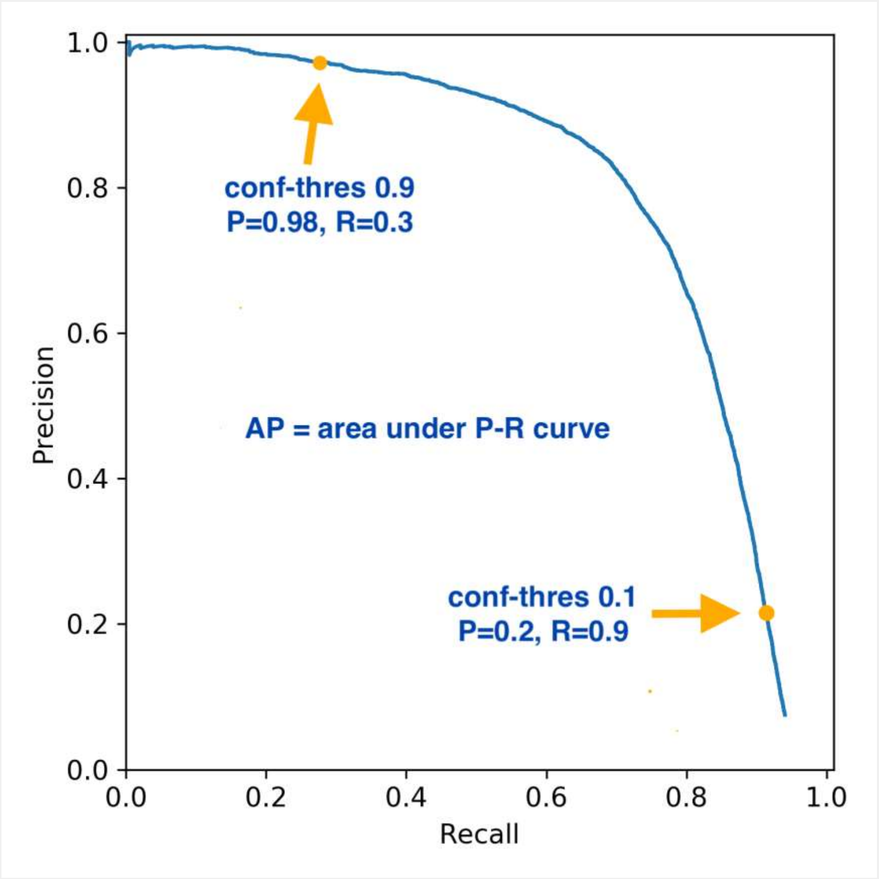
•A higher AUC-PR means better overall performance across all thresholds.

◆ **Average Precision (AP)**

•A **summary metric** that averages precision at different recall levels.

•**Average Precision (AP)** is a **step-wise approximation** of the **area under the precision-recall curve (AUC-PR)**.

◆ **ROC AUC:** https://scikit-learn.org/1.0/auto_examples/model_selection/plot_roc.html



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Classification Metrics: Generalization to multi-class

- **One-vs-Rest (OvR)**: Most common strategy — treat each class as positive vs. all others as negative to compute metrics like precision, recall, F1.

- **Macro average**: Compute metric independently for each class, then take the **unweighted mean**.
  → Treats all classes **equally**, even rare ones.

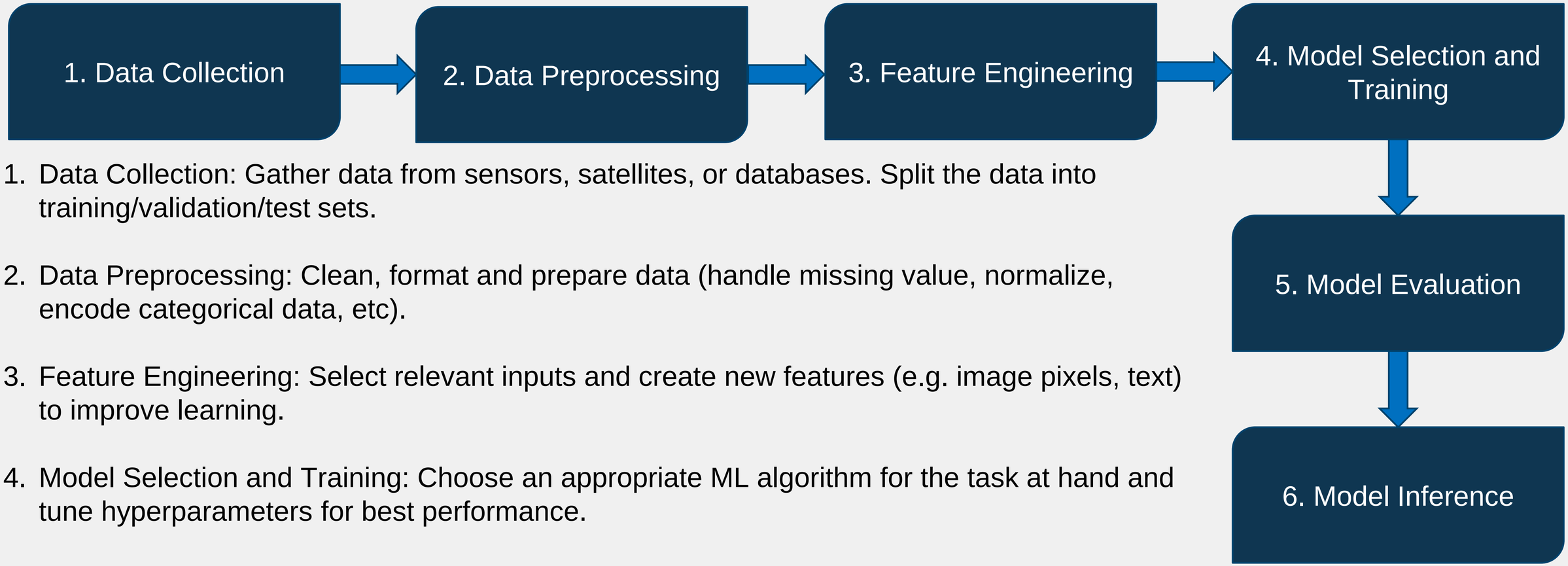
- **Micro average**: Aggregate total true positives, false positives, and false negatives over all classes, then compute the metric.
  → Gives **more weight to common classes**.

- **Weighted average**: Like macro, but each class's metric is **weighted by its support** (number of true instances).
  → Balances class imbalance while keeping per-class perspective.

- **Additional resources**: https://www.evidentlyai.com/classification-metrics/multi-class-metrics

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The Machine Learning Pipeline



1. Data Collection: Gather data from sensors, satellites, or databases. Split the data into training/validation/test sets.

2. Data Preprocessing: Clean, format and prepare data (handle missing value, normalize, encode categorical data, etc).

3. Feature Engineering: Select relevant inputs and create new features (e.g. image pixels, text) to improve learning.

4. Model Selection and Training: Choose an appropriate ML algorithm for the task at hand and tune hyperparameters for best performance.

5. Model Evaluation: Assess the model's accuracy and generalization using test data.

6. Model Inference: Use the trained model to make prediction and gain insights from new data.

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03

Overview of Popular ML Models



CONFIDENTIAL

(Penalized) Linear Regression

◆ **What it does**

•Linear Regression models the relationship between input features **X** and a continuous output **y**.

•The model tries to find the **best-fitting line** (or hyperplane) through the data.

◆ **Prediction Formula**

$$\hat{y} = WX + B$$

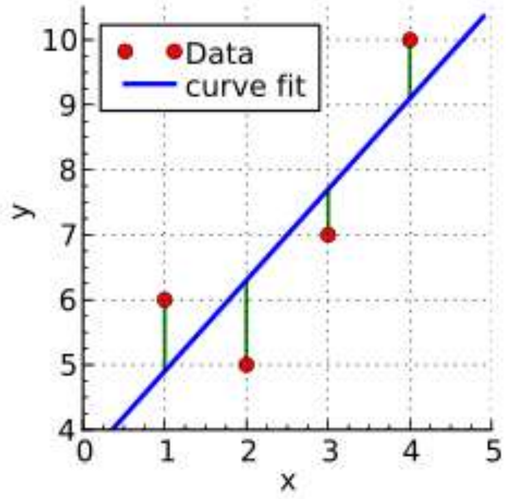
•Where:

- X = input features in vector format
- W = **weights** in vector format
- B = **biases** in vector format
- $\hat{y}$ = predicted output or target

◆ **Loss to minimize**

$$\mathcal{L} = \sum_{i=1}^{N}(y_i - f(x_i, w, b))^2 + \alpha \sum_{j=1}^{D} w_j^2 + \beta \sum_{j=1}^{D} |w_j|$$

- **Lasso** ($\alpha = 0, \beta \neq 0$), **Ridge** ($\alpha \neq 0, \beta = 0$) and **ElasticNet** ($\alpha \neq 0, \beta \neq 0$)
- Same kind of regularization techniques also used for neural networks
- **Additional resources:** https://www.geeksforgeeks.org/lasso-vs-ridge-vs-elastic-net-ml/



$$MAE = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}|$$

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y})^2$$

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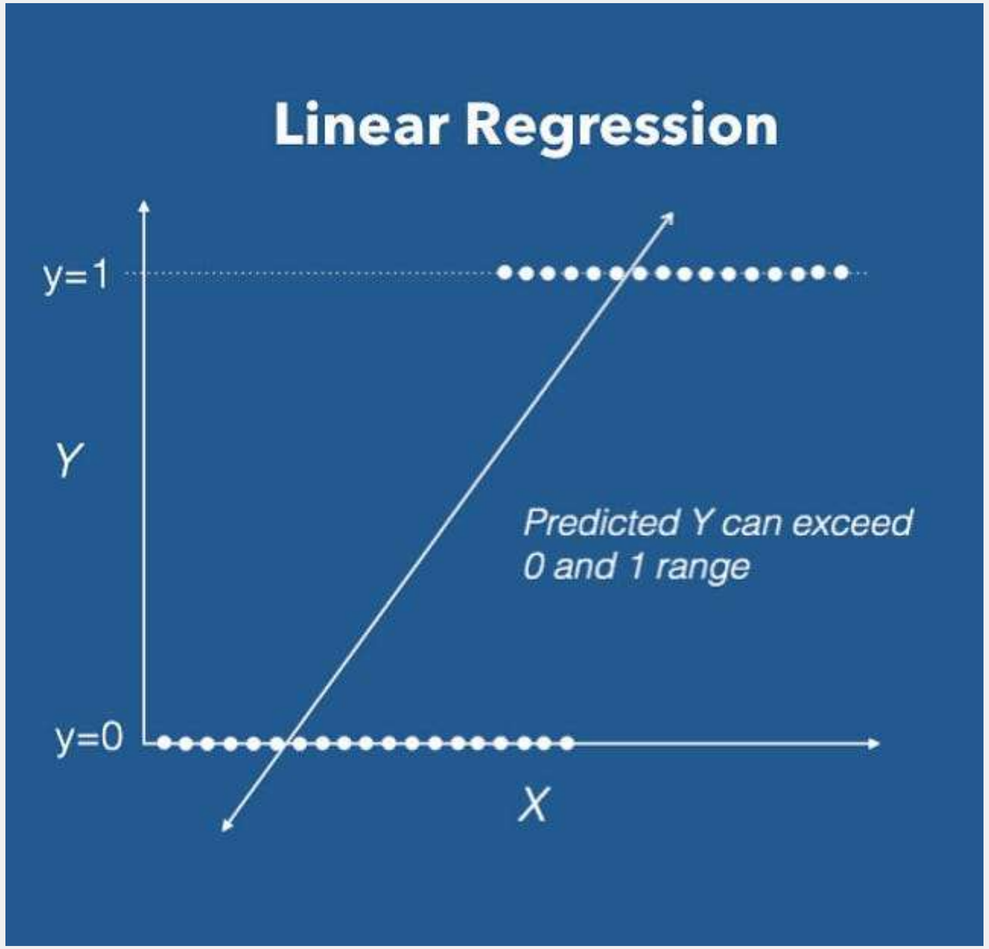
Where,

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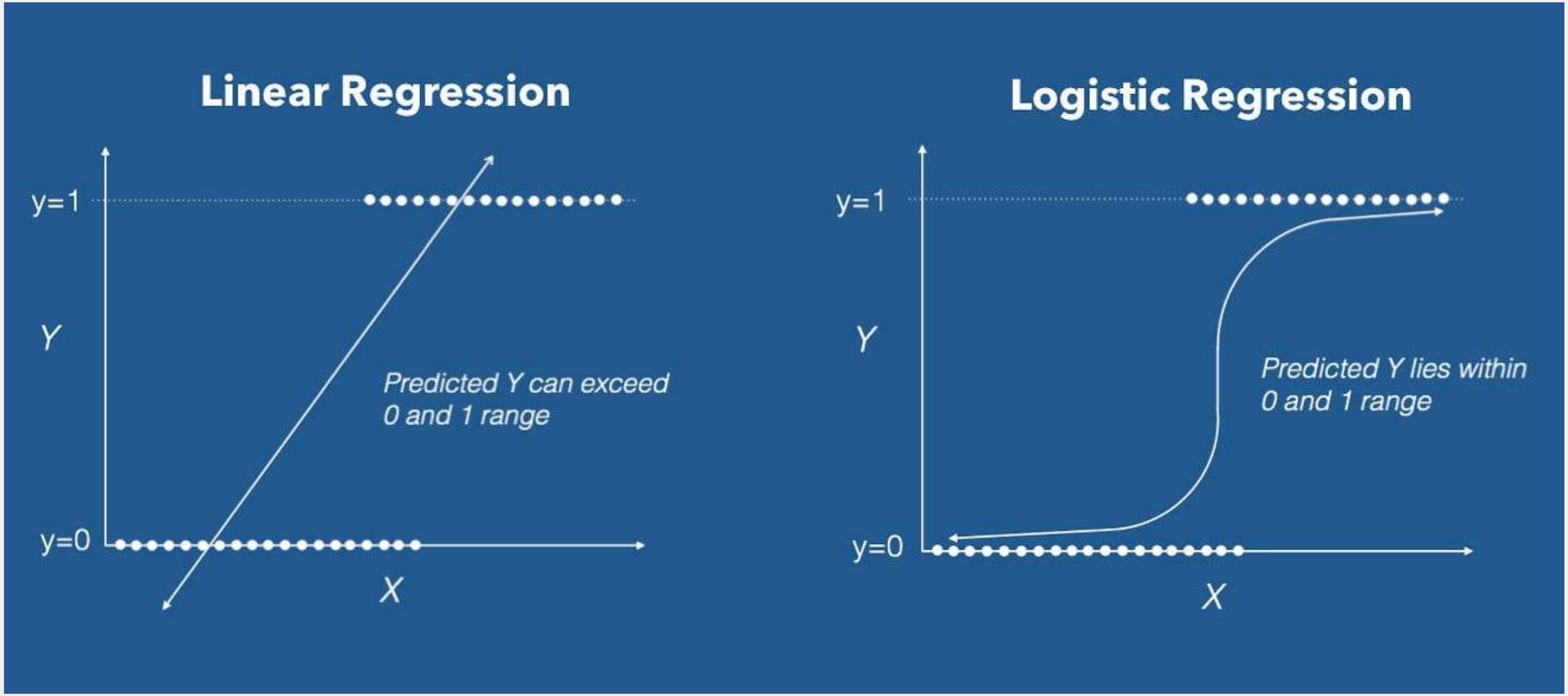
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Linear Model for Classification



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Linear Model for Classification



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Logistic Regression (single feature case)

◆ **What it does:** Logistic Regression models the probability that an input belongs to a specific class (usually binary: 0 or 1). It is a linear model for **classification**, not regression.

◆ **Prediction Formula**

$$\hat{y} = \sigma(w_1 x + b_1), \qquad \sigma(z) = \frac{1}{1 - e^{-z}}$$

The output is a **probability between 0 and 1**. You choose a threshold (typically 0.5) to make a class prediction.

◆ **Training the Model**

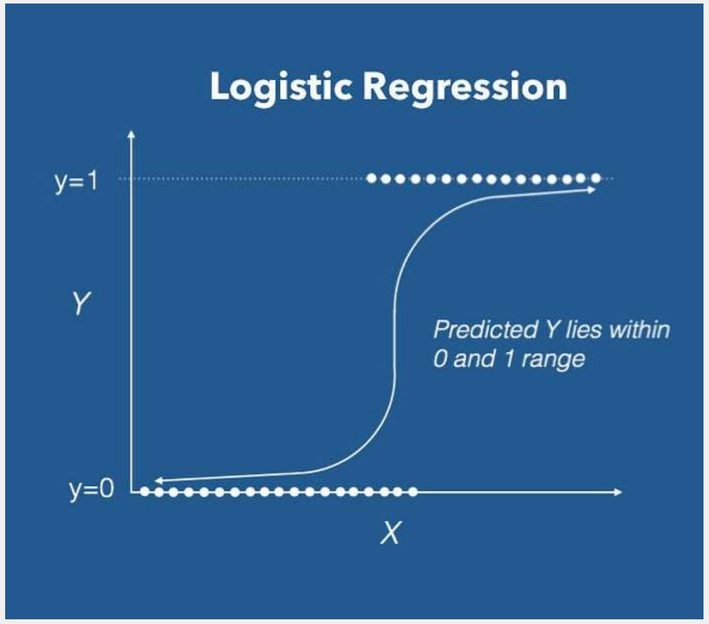
The model learns weights by **minimizing the log loss** (also called binary cross-entropy), often using gradient descent.

$$L_{BCE} = -\frac{1}{n}\sum_{i=1}^{n}(Y_i \cdot \log \hat{Y}_i + (1 - Y_i) \cdot \log (1 - \hat{Y}_i))$$

◆ **Outputs of the Model**

- Predicted probabilities for each class.
- Final class predictions (0 or 1).
- Performance metrics like
- **accuracy**, **precision**, **recall**, **F1-score**, etc.

🔹 Also possibility to penalize as for Lasso/Ridge/ElasticNet



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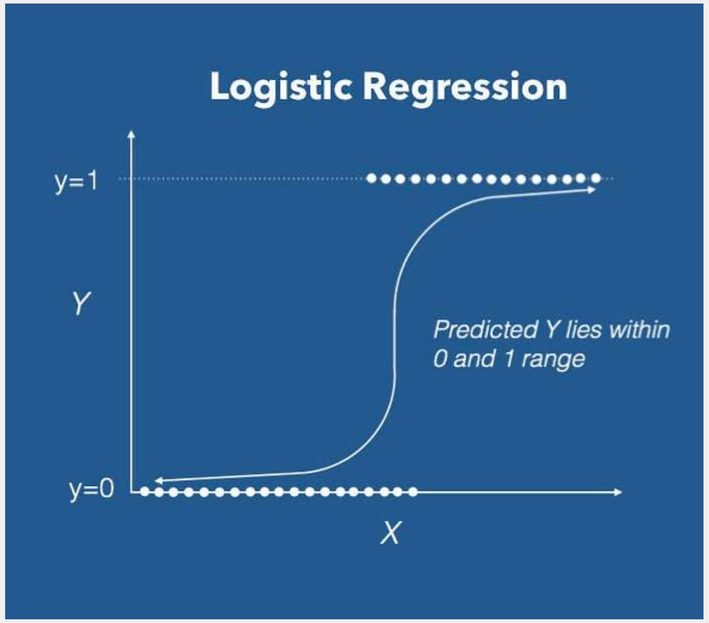
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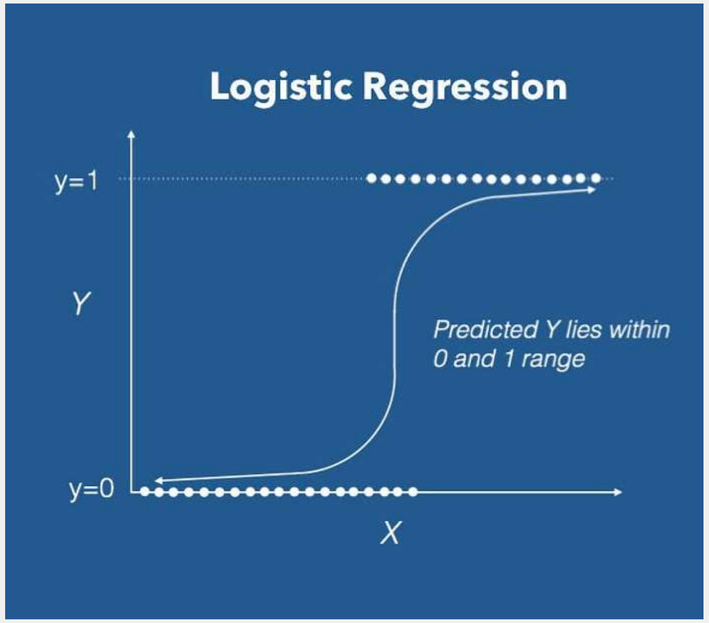
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k-Nearest Neighbors (KNN)

◆ **What it does**

k-NN is a simple, instance-based learning algorithm that makes predictions based on the labels of the *k* closest training examples in the feature space.

◆ **How it works**

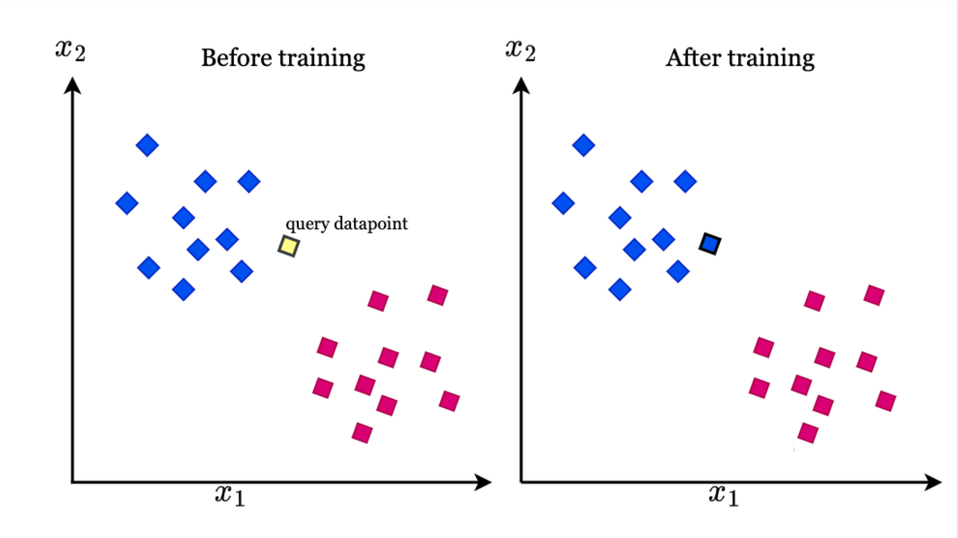
1. Compute the distance (e.g., Euclidean) between the input sample and all training samples.
2. Select the *k* nearest neighbors.
3. **For classification**: Predict the most common class among neighbors.
   For regression: Predict the average of neighbors' target values.

◆ **Key hyperparameters**

- **k**: Number of neighbors to consider.
- **Distance metric**: Euclidean, Manhattan, etc.

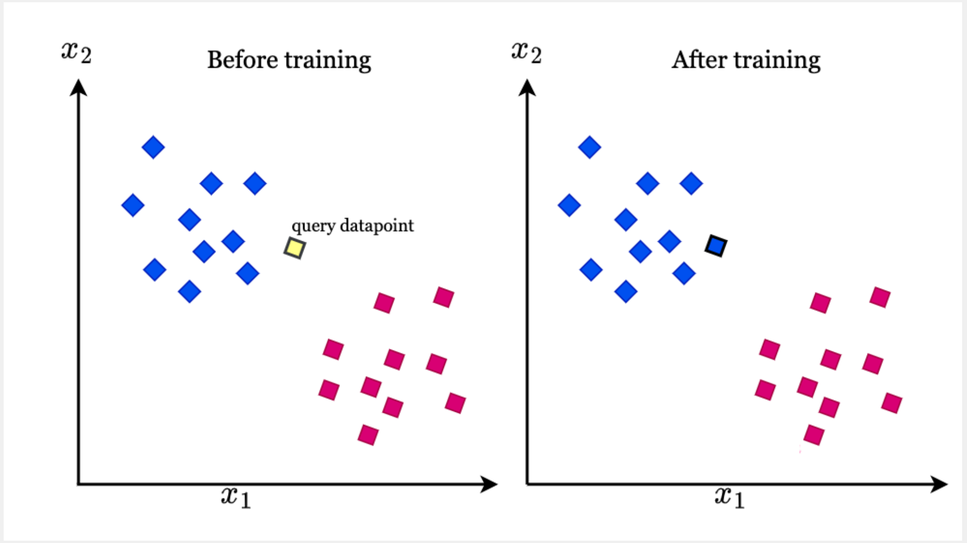
◆ **Pros and Cons**

✓ No training phase (lazy learner)
✗ Slow at prediction time (especially on large datasets)
✗ Sensitive to feature scaling and irrelevant features



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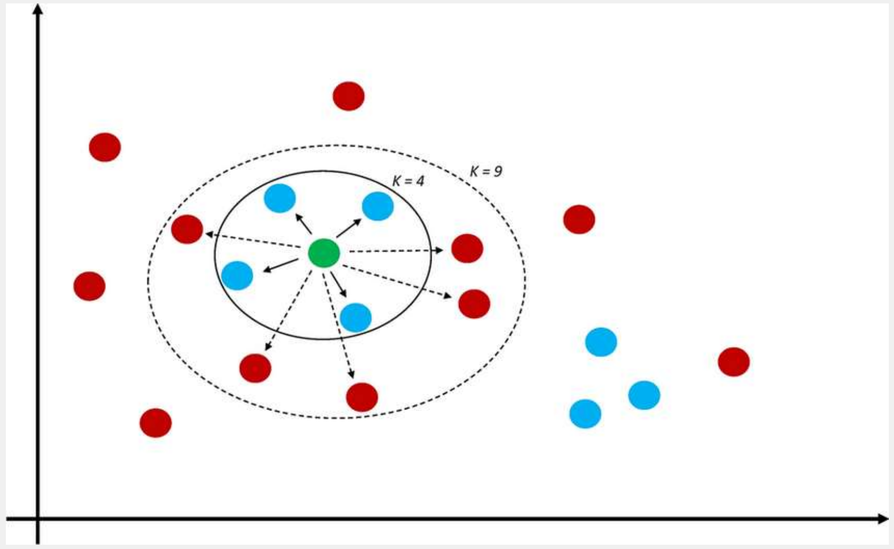


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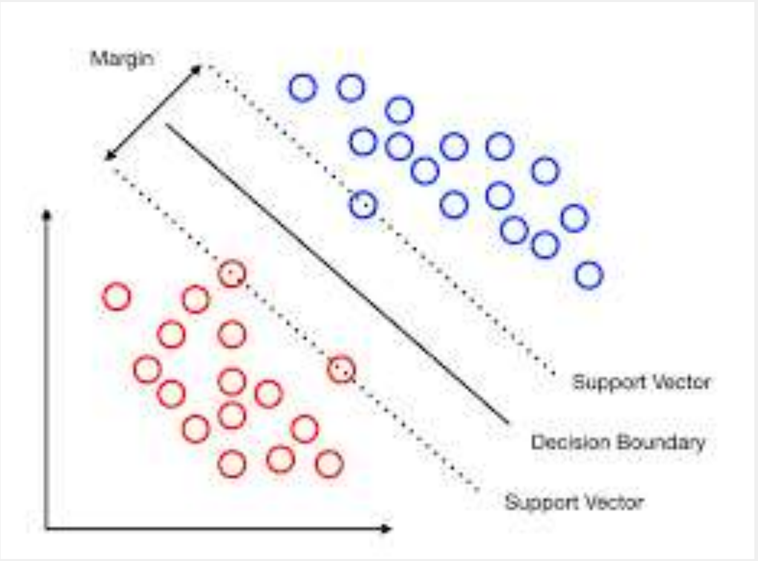
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Support Vector Machines (SVM)



◆ **What it does**

SVM aims to find the optimal boundary (hyperplane) that separates classes with the **maximum margin**.

- **Classification (SVC)**: Finds the hyperplane that best separates classes.
- **Regression (SVR)**: https://en.wikipedia.org/wiki/Support_vector_machine#Regression

◆ **Key Concepts**

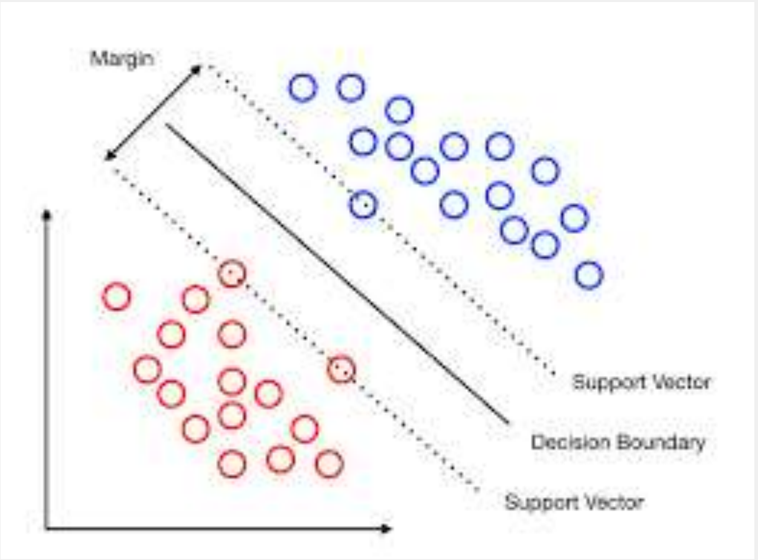
- **Support Vectors**: Data points closest to the decision boundary — critical for defining the model.
- **Kernel Trick**: Enables learning non-linear patterns by implicitly mapping data to higher dimensions (e.g., RBF, polynomial).
- **Margin**: Distance between the hyperplane and the closest data points from each class.

◆ **Pros and Cons**

✅ Effective in high-dimensional spaces
✅ Works well with clear margin of separation
❌ Computationally expensive on large datasets
❌ Requires careful tuning of parameters (kernel, C, epsilon)

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Support Vector Machines (SVM)



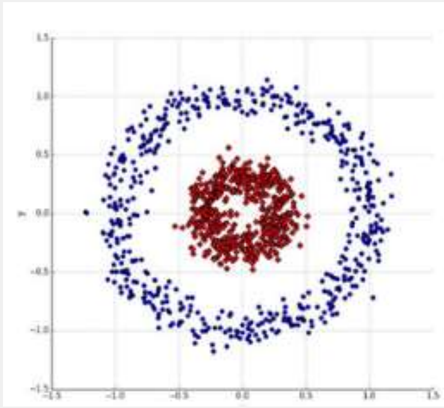
◆ **What it does**

SVM aims to find the optimal boundary (hyperplane) that separates classes with the **maximum margin**.

- **Classification (SVC)**: Finds the hyperplane that best separates classes.
- **Regression (SVR)**: https://en.wikipedia.org/wiki/Support_vector_machine#Regression

◆ **Key Concepts**

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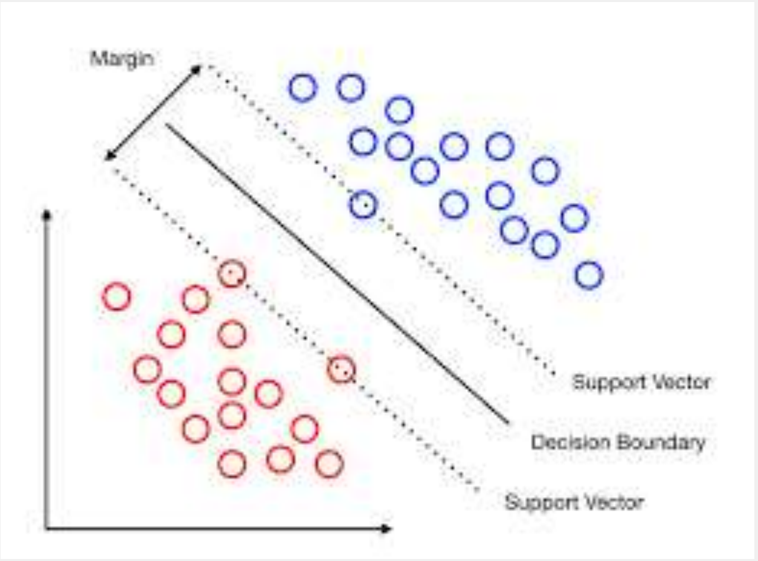


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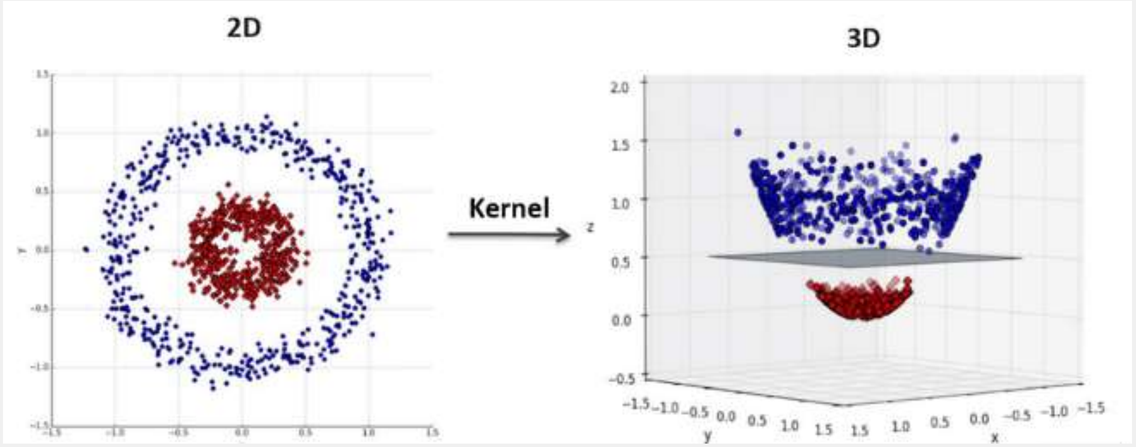
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Decision Trees

◆ **What it does**

- A Decision Tree splits the data into regions by asking a sequence of questions (e.g., *"is feature A < 3.5?"*)
- It learns simple decision rules from the features to predict an outcome.

◆ **How it works**

- Starting from the root node, it chooses the best feature and threshold to **split** the data to reduce "impurity".

◆ **Differences between classification & regression:**

•**Classification:**

- Goal: Predict a class label.
- Split criterion: **Gini impurity** or **entropy (information gain)**
- Output: Most common class in the leaf.

•**Regression:**

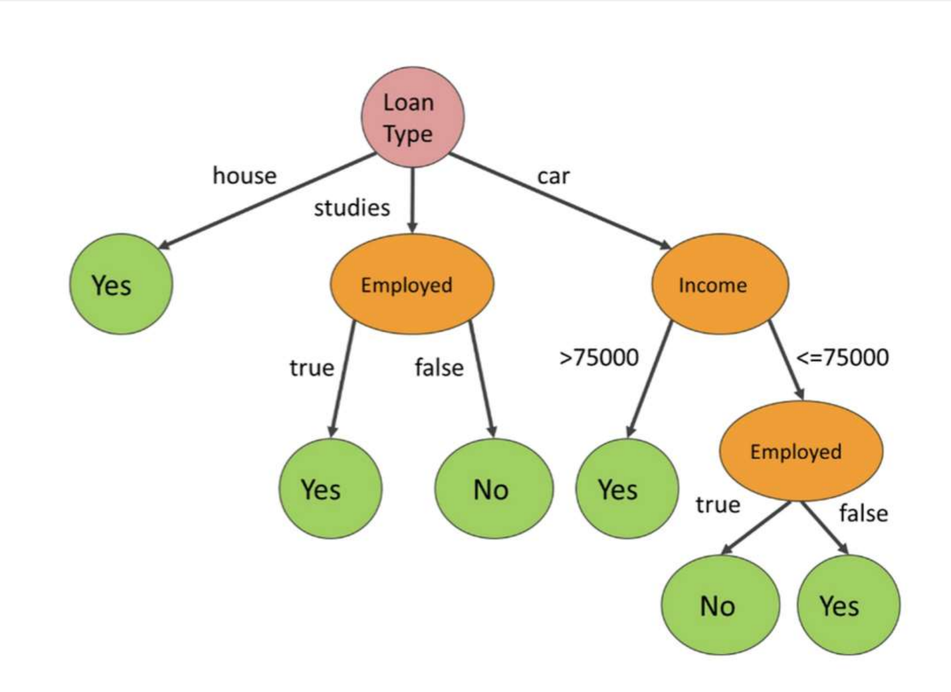
- Goal: Predict a continuous value.
- Split criterion: **Mean Squared Error (MSE)**
- Output: Average target value in the leaf.

◆ **Pros & Cons**

•Easy to interpret and visualize

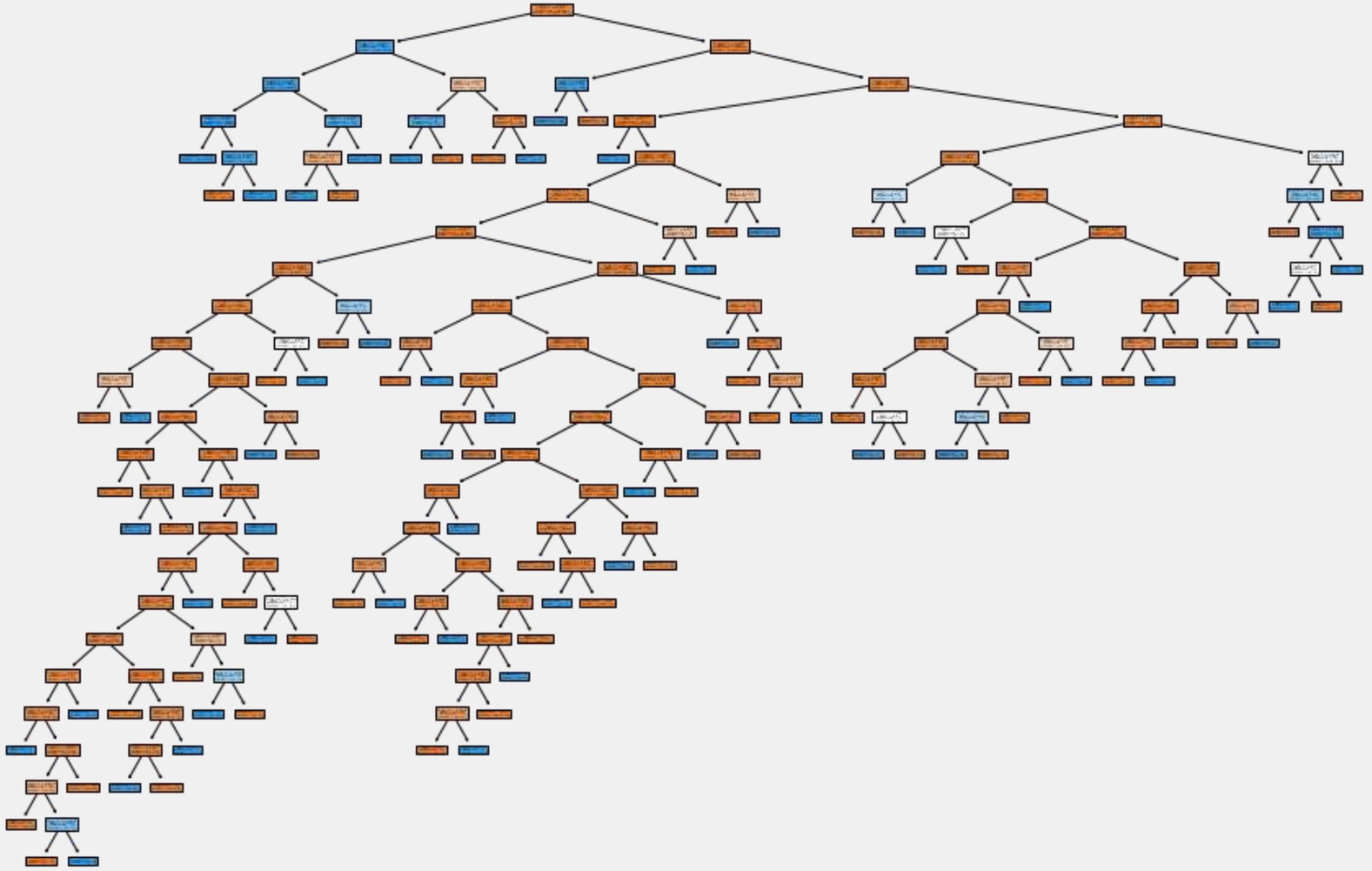
•Handles both numerical and categorical data

– Prone to overfitting without regularization (e.g., max depth, min samples)



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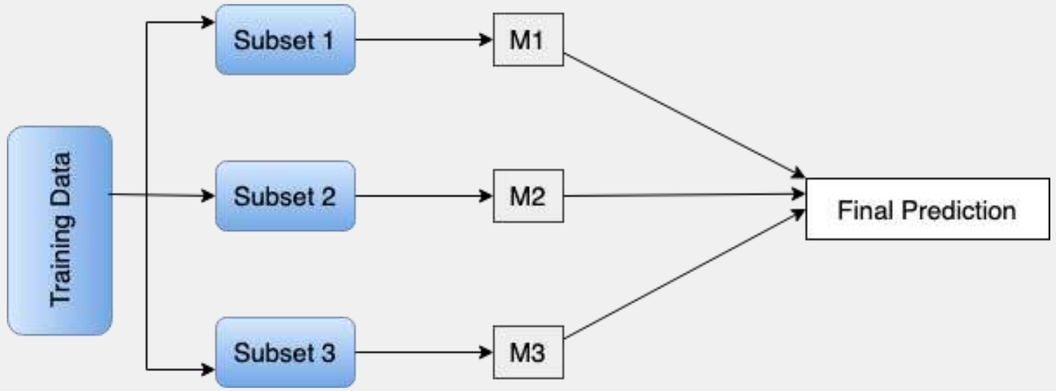
Decision Trees



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Ensemble Learning: Bagging

Bagging, also known as **B**ootstrap **Agg**regation, is an ensemble learning technique that combines the benefits of bootstrapping and aggregation to yield a stable model and improve the prediction performance of an ML model.



Pros	Cons
Improves overall accuracy	Can be computationally expensive to train many weak models in parallel
Reduces overall variance	High number of weak models may reduce model interpretability
Increases models' robustness to noise in the data	

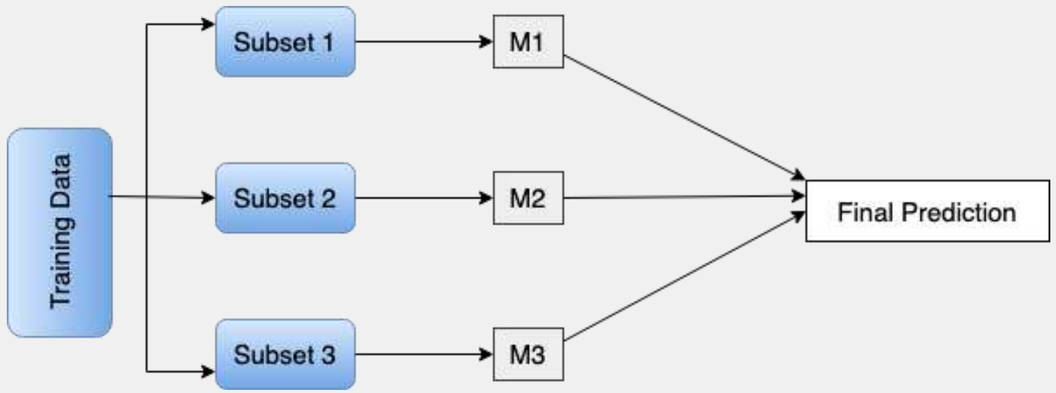
The main idea behind bagging is to reduce the variance in a dataset (aka reducing overfitting), ensuring that the model is robust and not influenced by specific samples in the dataset.

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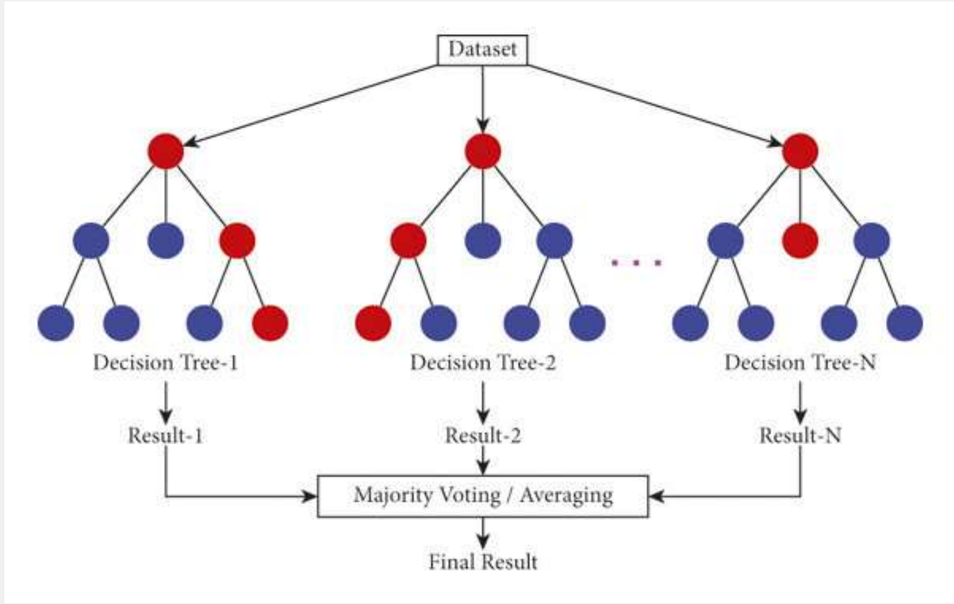
3. Overview of Popular ML Models

Ensemble Learning: Bagging ➔ Random Forest

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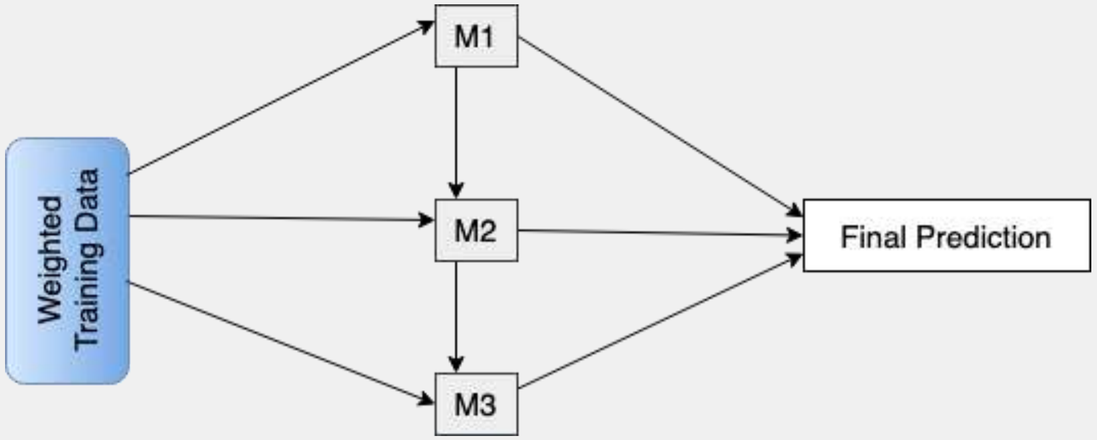
The main idea behind bagging is to reduce the variance in a dataset (aka reducing overfitting), ensuring that the model is robust and not influenced by specific samples in the dataset.

Example of Bagging is Random Forest Decision from Decision Trees.

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Ensemble Learning: Boosting

In boosting, we train a sequence of models, each one aiming to correct the errors of its predecessors.
Each model is trained on a weighted dataset, where instances misclassified by previous models are given **higher weights**, so that subsequent models focus more on the harder cases.
This sequential process continues until a predefined number of models is reached or another stopping criterion is met.



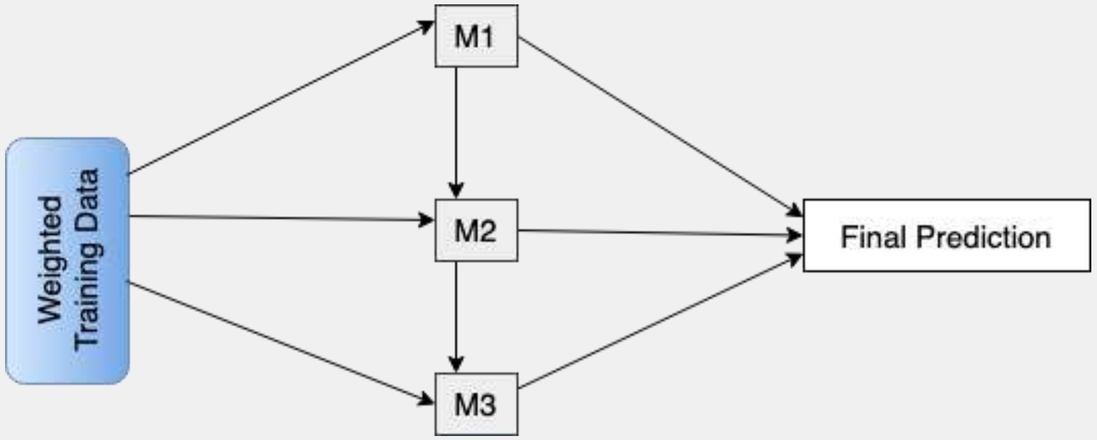
Pros	Cons
Improves overall accuracy	Can be computationally expensive
Reduces overall bias by improving on the weakness of the previous model	Sensitive to noisy data
	Model dependency may allow for replication of errors

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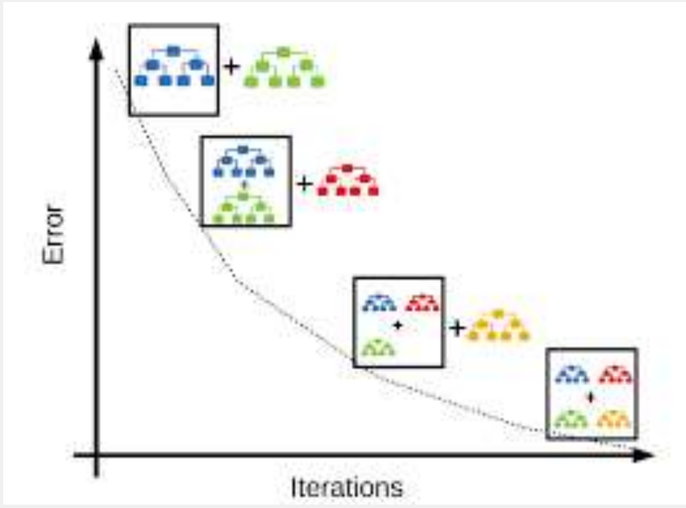
3. Overview of Popular ML Models

Ensemble Learning: Boosting ➔ Boosted Trees (XGBoost, LightGBM, …)

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Reduces overall bias by improving on the weakness of the previous model	Sensitive to noisy data
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Example of Boosting are XGBoost, LightGBM from Decision Trees.

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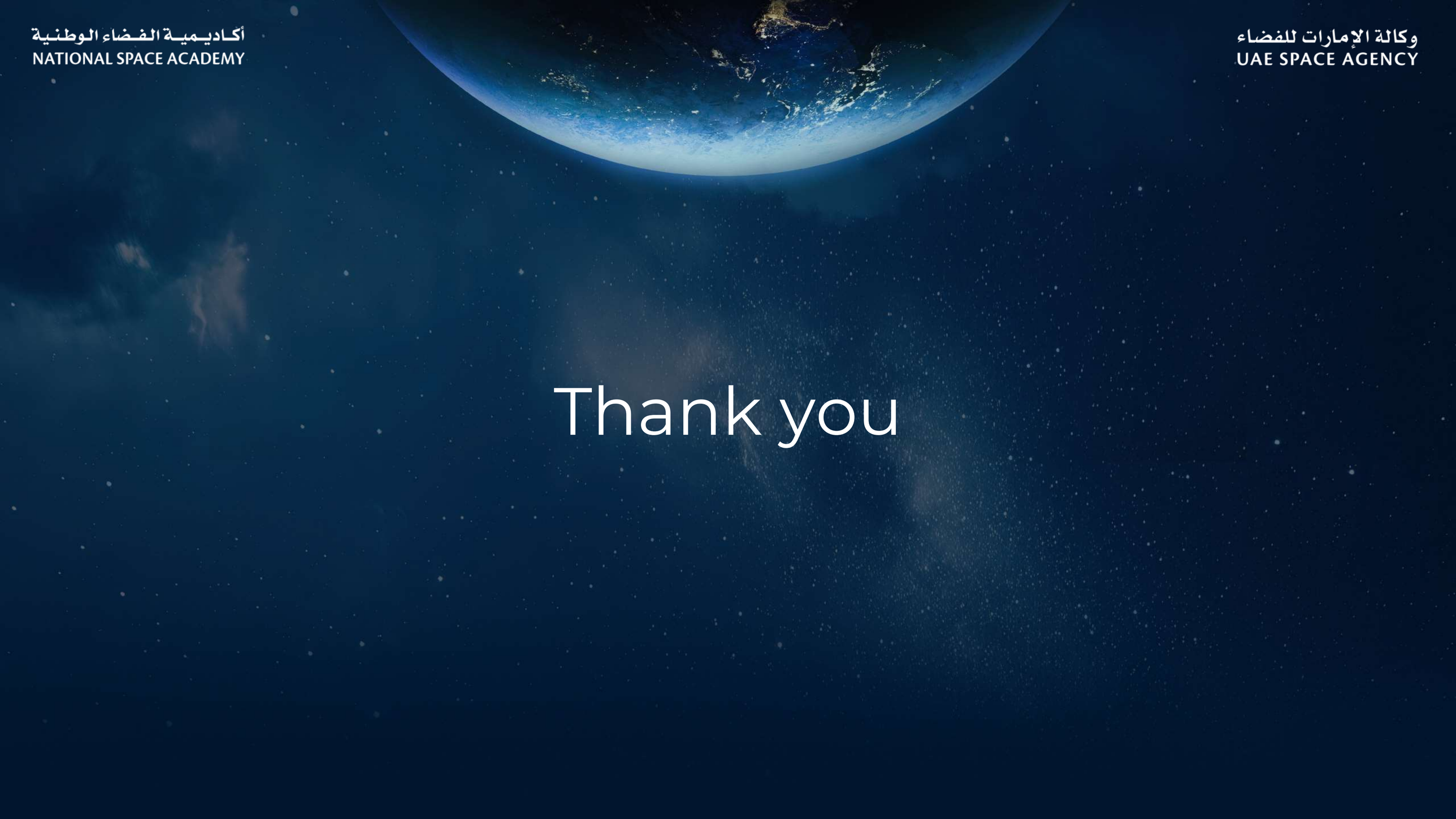
Additional Resources

🔹 **Handling Categorical Variables**
•**Ordinal Encoding**: Converts categories to ordered integers (useful for ordinal variables).
🔗 https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.OrdinalEncoder.html
•**One-Hot Encoding (OHE)**: Creates binary columns for each category (useful for nominal variables).
🔗 https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.OneHotEncoder.html
• 🔹 **Feature Scaling**
•**StandardScaler**: Removes the mean and scales to unit variance (important for distance-based models).
🔗 https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.StandardScaler.html
•**MinMaxScaler**: Scales features to a [0, 1] range.
🔗 https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.MinMaxScaler.html
🔹 **Feature Generation / Combination**
•Create new features (e.g. ratios, interaction terms) to add signal.
•Can also use **PolynomialFeatures** for adding interactions and squared terms.
🔗 https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.PolynomialFeatures.html
🔹 **ColumnTransformer & Pipelines**
•Combine preprocessing steps for different columns in one reusable object.
🔗 https://scikit-learn.org/stable/modules/generated/sklearn.compose.ColumnTransformer.html
🔗 https://scikit-learn.org/stable/modules/generated/sklearn.pipeline.Pipeline.html
🔹 **Model Ensembling at the Preprocessing Level**
•Use **stacking** to combine predictions from multiple models as new features.
🔗 https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.StackingClassifier.html
🔗 https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.StackingRegressor.html

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Thank you